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TEA SENSE: A Spectroscopic System for Estimating Moisture and Quality Parameters in Black Tea

Final Year Project Proposal

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ABSTRACT

Black-tea quality depends on moisture, polyphenols, the theaflavin to thearubigin ratio (TF:TR), caffeine and sugar, and safety depends on residues such as glyphosate. Each is measured by a separate laboratory method; moisture alone takes about six hours by oven-drying to constant mass, while the others need extraction, spectrophotometry, HPLC or LC-MS/MS. Results therefore arrive after processing and trading decisions are made.

This project proposes and evaluates a Near-Infrared (NIR) spectroscopy and Artificial Intelligence (AI) framework in which one portable NIR scan, taking seconds and destroying nothing, feeds parameter-specific prediction models. The framework is built and validated experimentally on moisture, using oven-drying as the reference, by comparing spectral preprocessing (SNV, MSC, Savitzky-Golay), wavelength selection (CARS) and models (PLSR, SVR, Random Forest and an optional neural network) on held-out and source-held-out Sri Lankan black-tea samples, and by measuring the time saved. A structured review of 45 studies rates which further parameters the same framework can realistically estimate.

The literature shows strong NIR evidence for moisture, caffeine, TF:TR and polyphenols, emerging evidence for added sugar, and no credible NIR route for trace glyphosate. A preliminary proof of concept on a public NIR moisture dataset reached an RMSEP of 1.04% (R2p 0.989) with SVR and confirmed the leakage-free pipeline. The expected outcome is a validated moisture model that cuts measurement time from about six hours to under one minute, a reproducible multi-parameter framework and an evidence-based feasibility map for the remaining parameters.

Keywords: Near-Infrared Spectroscopy (NIR), Tea Moisture, Tea Quality Parameters, Artificial Intelligence, Machine Learning, Chemometrics, Spectral Preprocessing, Quality Control

DECLARATION AND GENERATIVE AI DISCLOSURE

Signature: Saad_____ Date: _____

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LIST OF ABBREVIATIONS

Abbreviation	Meaning
AI	Artificial Intelligence
ANN	Artificial Neural Network
BCS	BCS, The Chartered Institute for IT
BiGRU	Bidirectional Gated Recurrent Unit
CARS	Competitive Adaptive Reweighted Sampling
CNN	Convolutional Neural Network
CTC	Crush Tear Curl
CV	Cross-Validation
ELM	Extreme Learning Machine
FT-NIR	Fourier-Transform Near-Infrared
FTIR-ATR	Fourier-Transform Infrared, Attenuated Total Reflectance
HPLC	High-Performance Liquid Chromatography
iPLS	Interval Partial Least Squares
ISO	International Organization for Standardization
KS	Kennard-Stone
LC-MS/MS	Liquid Chromatography with Tandem Mass Spectrometry
LIBS	Laser-Induced Breakdown Spectroscopy
LOQ	Limit of Quantification
LOSO	Leave-One-Source-Out
MSC	Multiplicative Scatter Correction
NIR	Near-Infrared
PCA	Principal Component Analysis
PLSR	Partial Least Squares Regression
R2	Coefficient of determination (R2p on the prediction set)
RF	Random Forest
RMSEC / RMSECV / RMSEP	Root Mean Square Error of Calibration / Cross-Validation / Prediction
RPD	Ratio of Performance to Deviation

Abbreviation	Meaning
RQ	Research Question
SDG	Sustainable Development Goal
SERS	Surface-Enhanced Raman Spectroscopy
SG	Savitzky-Golay
SNV	Standard Normal Variate
SPA	Successive Projections Algorithm
SUS	System Usability Scale
SVR	Support Vector Regression
TF / TR	Theaflavins / Thearubigins
THz	Terahertz
TRI	Tea Research Institute of Sri Lanka
UV-Vis	Ultraviolet-Visible
VIP	Variable Importance in Projection

GLOSSARY

Term	Meaning
Calibration set	Samples used to build and tune a model; never used for final evaluation.
Chemometrics	Statistical and mathematical methods for extracting chemical information from measurement data such as spectra.
Diffuse reflectance	Light scattered back from inside a solid sample, which carries information about its composition.
Kennard-Stone split	A data split that selects calibration samples to cover the spectral space evenly.
Leave-one-source-out	Validation in which all samples from one estate or factory are held out and predicted by a model trained on the rest.
Made tea	Finished black tea after drying and sorting.
Reference method	The accepted laboratory method that provides ground-truth values, such as oven-drying for moisture.
TF:TR ratio	The ratio of theaflavins to thearubigins, a black-tea fermentation and quality indicator.
Wavelength selection	Choosing the most informative wavelengths so a model uses fewer, more relevant variables.

CHAPTER 1: INTRODUCTION

1.1 Chapter overview

This chapter introduces the research domain of rapid tea quality assessment using near-infrared (NIR) spectroscopy and artificial intelligence (AI). It explains why tea parameters such as moisture, polyphenols, the TF:TR ratio, caffeine, sugar and glyphosate matter, why conventional laboratory methods are slow, and how NIR and AI could shorten the measurement cycle. It then defines the problem, motivation, research gaps, contributions, challenges, research questions, aim, objectives, scope and expected outcomes, and leads into the literature review in Chapter 2.

1.2 Problem background

1.2.1 Research domain context

Tea quality is governed by physical, chemical and safety parameters. Moisture controls shelf life, mould risk and saleable weight; total polyphenols and catechins shape astringency; theaflavins (TF) and thearubigins (TR), formed during fermentation, give black tea its brightness, briskness and colour (Roberts and Smith, 1961); caffeine affects briskness and labelling; added sugar is an adulterant; and glyphosate residues are a market-access concern.

Sri Lanka exported 257.44 million kg of tea in 2025, earning about US$1.51 billion, up from 245.79 million kg in 2024 (Daily FT, 2026). At this scale, the speed and reliability of quality testing have direct commercial value, and integrity problems have real consequences: in 2018 the Sri Lanka Tea Board found 53 of 80 inspected factories mixing sugar or glucose into tea to darken it (Daily Mirror, 2018).

Each parameter is measured by its own laboratory reference method (Table 1). Moisture is determined by drying a sample at about 103 C to constant mass (ISO 1573), which takes about six hours in the author's laboratory procedure. The other parameters need solvent extraction, spectrophotometry, HPLC or LC-MS/MS. These methods are accurate but slow, destructive and laboratory-bound, so results usually arrive after drying, grading or trading decisions have been made.

Table 1: Conventional reference methods for black-tea parameters

Parameter	Typical level	Reference method	Time and access
Moisture	About 2.5 to 7% in made tea	Oven-drying to constant mass at 103 C (ISO 1573)	About 6 h per batch; destructive
Total polyphenols	Percent level	Folin-Ciocalteu colorimetry (ISO 14502-1)	Hours including extraction; reagents
TF and TR (TF:TR)	Percent level	Roberts and Smith (1961) spectrophotometry	Hours; solvent partitioning
Caffeine	Percent level	HPLC	About 15 min run per sample after preparation (Rehman et al., 2020)
Sugars	Natural traces; adulteration up to several percent	HPLC with refractive-index detection (Piyasena et al., 2022)	Laboratory only
Glyphosate	Trace (mg/kg)	LC-MS/MS; LOQ 1.8 mg/kg (Samanthy et al., 2021)	Specialist laboratory

Times other than moisture are indicative and will be confirmed with a tea laboratory. The research domain of this study is therefore the use of spectroscopy and AI to estimate several tea parameters rapidly, with moisture as the first, experimentally validated parameter.

1.2.2 Specific technological context

NIR spectroscopy (about 780 to 2500 nm) records overtone and combination absorptions of O-H, C-H and N-H bonds. Water, polyphenols, caffeine and sugars all contain these bonds, so one NIR spectrum carries information about several components, and diffuse reflectance lets it read solid tea without preparation (Hazarika et al., 2017). Water has strong bands near 970, 1450 and 1940 nm, which makes moisture the most direct NIR target.

Raw spectra contain scatter from particle size and packing, baseline drift and noise, so preprocessing such as Standard Normal Variate (SNV), Multiplicative Scatter Correction (MSC) and Savitzky-Golay (SG) smoothing and derivatives is normally applied (Qiu et al., 2026). Wavelength-selection methods such as Competitive Adaptive Reweighted Sampling (CARS) keep the most informative bands (Hazarika et al., 2024). Partial Least Squares Regression (PLSR) is the standard chemometric model, while machine learning (SVR,

Random Forest, Extreme Learning Machines) and deep learning can capture non-linear relationships (Zou et al., 2022; Dong et al., 2017; Tao et al., 2026).

NIR has limits. It reads components present at percent levels well, but trace residues have no direct NIR signature; metals, for example, are predicted only through correlations with the tea matrix (Revilla et al., 2024). Other technologies (UV-Vis, FTIR, terahertz, LIBS, SERS) are considered where a parameter needs them, but NIR is the primary sensing technology for this study.

1.2.3 Practical and real-world context

Factories, estates, brokers, exporters, the Tea Board and laboratories all need quality information, but samples must usually be carried to a laboratory, prepared and analysed. A portable NIR system would not replace laboratory testing; it would act as a rapid screening layer, giving a moisture value in seconds and flagging samples that need laboratory confirmation. Real environments introduce variation in particle size, moisture distribution, temperature, cultivar, origin, processing stage, lighting and handling, so a model that works in the laboratory may not work in a factory. This research therefore evaluates the approach on real Sri Lankan tea samples and under practical conditions.

1.3 Problem definition

Tea quality assessment requires several parameters, each measured by a slow, destructive, laboratory-bound method. NIR spectroscopy with AI has shown strong results for individual tea parameters, but the evidence comes almost entirely from single cultivars, sites and seasons outside Sri Lanka, rarely reports time savings or field accuracy, and does not establish which parameters one portable NIR system can realistically cover. The research does not assume that one technology can measure every parameter. It builds and validates the framework on moisture, where an in-house reference method exists, and assesses the other parameters through structured evidence.

1.3.1 Problem statement

There is a need to evaluate whether near-infrared spectroscopy combined with chemometric and AI-based modelling can estimate black-tea quality parameters, beginning with moisture, accurately and rapidly enough for screening under practical conditions, and to establish which further parameters such a framework can realistically estimate.

1.4 Research motivation

Practical motivation. The industry needs results while decisions can still change: a dryer operator needs moisture during the batch, and a broker or the Tea Board needs to screen many lots quickly for adulteration. Replacing about six hours of oven-drying with a scan of under a minute would change how often and where testing can happen.

Technological motivation. Portable 900 to 1700 nm spectrometers and open-source machine learning now make low-cost NIR systems feasible (Li et al., 2023; Tang et al., 2025). The academic motivation is to establish which preprocessing, wavelength-selection and modelling choices work for Sri Lankan black tea, whether AI models genuinely beat chemometrics at realistic sample sizes, and which parameters NIR can and cannot serve, rather than assuming one sensor measures everything.

1.5 Research gap analysis

The gap is not the absence of NIR in tea research; NIR has been applied to moisture (Zou et al., 2022; Lan et al., 2022; Tao et al., 2026), caffeine (Rehman et al., 2020; Zareef et al., 2018), TF:TR (Dong et al., 2017), polyphenols (Liu et al., 2024; Xu et al., 2026) and added sugar (Luo et al., 2023). The gap is that this evidence is fragmented and local validation is missing. Five gaps follow from the literature, which Chapter 2 examines in detail.

Gap 1: No validated NIR moisture model for Sri Lankan black tea. Every moisture study reviewed used a single cultivar, site or season, and none used Ceylon black tea. **Gap 2: Preprocessing and wavelength-selection effects are task-specific.** Preprocessing raised class separability from 53.63 to 93.20 in one study (Das et al., 2020), and selection with few wavelengths worked for CTC tea (Hazarika et al., 2024), but the best strategy cannot be transferred from another parameter. **Gap 3: Calibration does not guarantee generalisation.** Validation accuracy fell below calibration accuracy in Revilla et al. (2024), and model universality is the main unresolved challenge (Qiu et al., 2026). **Gap 4: Time savings and field robustness are rarely measured. Gap 5: No consolidated assessment of which tea parameters one NIR framework can cover,** including a clear position on trace residues such as glyphosate.

Table 2: Research gap summary

Research gap	Evidence from literature	Planned response	RQ	Objective
G1 No validated moisture model for Sri Lankan black tea	Single-site moisture studies (Zou 2022; Lan 2022; Tao 2026; Kang 2026)	Collect paired Sri Lankan data; validate on held-out samples	RQ1	O2, O6
G2 Preprocessing and selection effects unknown for this task	Das 2020; Hazarika 2024; Xu 2026	Compare SNV, MSC, SG; CARS vs SPA/iPLS	RQ2	O3, O4
G3 Calibration does not guarantee generalisation; AI vs PLSR unresolved	Revilla 2024; Qiu 2026; Zou 2022; Luo 2023	Identical splits; locked test; leave-one-source-out; paired tests	RQ3	O5, O6
G4 Time saving and field robustness unmeasured	Qiu 2026; Tao 2026	Time oven vs NIR; field test with oven subset	RQ4	O6, O7
G5 No consolidated multi-parameter feasibility evidence	Bai 2022 (uneven accuracy across parameters); Samanthy 2021 (trace glyphosate)	Graded feasibility matrix; extensible model registry	RQ5	O1, O8

1.6 Contribution to the body of knowledge

1.6.1 Research-domain contribution

The study contributes comparative, held-out evidence on how preprocessing, wavelength selection and model choice (chemometric versus machine learning) affect NIR moisture estimation in Sri Lankan black tea, including cross-source generalisation. It also contributes an evidence-graded synthesis of which tea parameters NIR can estimate, with a reasoned negative position on trace glyphosate.

1.6.2 Problem-domain contribution

The practical contribution is a measured comparison of time and accuracy between oven-drying and NIR screening, a reproducible open-source pipeline and prototype, and a roadmap for adding TF:TR, caffeine, polyphenol and sugar models when reference data are available.

1.7 Research challenges

Sample variability: cultivar, origin, grade, particle size and storage change spectra. **Narrow moisture range:** made tea sits at about 2.5 to 7% moisture, which makes R2 look poor even when error is small, so drying-stage and conditioned samples will widen the range. **Reference consistency:** model error cannot be lower than reference error, so oven-drying is done in triplicate under a written procedure. **Spectral noise and drift:** addressed through preprocessing, references and control samples. **Missing reference methods for other parameters:** only oven and balance are available in-house, so other parameters rely on evidence or a partner laboratory.

1.8 Research questions

RQ1

To what extent can NIR spectral measurements predict the moisture content of Sri Lankan black-tea samples when compared with oven-drying reference measurements on unseen samples?

RQ2

How do spectral preprocessing (SNV, MSC, Savitzky-Golay filtering) and wavelength selection (CARS) affect the predictive performance of tea moisture models?

RQ3

Do machine-learning models (SVR, Random Forest, and a neural network where data allow) improve on the PLSR chemometric baseline, and do the models generalise to samples from an unseen source?

RQ4

To what extent does the NIR approach reduce measurement time compared with oven-drying while remaining accurate under selected practical or field conditions?

RQ5

Which further parameters (polyphenols, TF:TR, caffeine, sugar, glyphosate) can the same NIR and AI framework estimate according to the evidence, and what would each require?

1.9 Research aim

The aim of this research is to design, develop and evaluate a near-infrared spectroscopy and AI framework for the rapid, non-destructive estimation of black-tea quality parameters, validated experimentally on moisture and assessed for extension to polyphenols, TF:TR, caffeine, sugar and glyphosate.

Reference moisture will be established by oven-drying and paired with NIR spectra from Sri Lankan black tea. Preprocessing, wavelength-selection and modelling options will be compared under one validation protocol, evaluated on held-out and source-held-out samples, and tested for speed and practicality in the field. In parallel, a structured evidence synthesis will rate the feasibility of the other parameters.

1.10 Research objectives

Table 3: Research objectives

ID	Objective	RQ
O1	Review and critically evaluate research on NIR spectroscopy, chemometrics and AI for tea quality parameters and their reference methods.	All
O2	Collect NIR spectra and oven-drying moisture values from Sri Lankan black-tea samples, retaining sealed subsamples for later reference analysis.	RQ1
O3	Investigate the effect of SNV, MSC, Savitzky-Golay smoothing and derivatives on moisture prediction.	RQ2
O4	Investigate wavelength selection (CARS against SPA or interval PLS) for moisture estimation.	RQ2
O5	Develop and compare PLSR, SVR, Random Forest and an optional neural network on identical data splits.	RQ3
O6	Evaluate models on held-out and source-held-out samples using RMSEP, R2, RPD and bias, and measure time per measurement.	RQ1, RQ3, RQ4
O7	Assess field feasibility with a prototype against an oven-checked subset.	RQ4
O8	Assess and design the extension to polyphenols, TF:TR, caffeine, sugar and glyphosate through a graded feasibility matrix and an extensible model registry.	RQ5

1.11 Scope, delimitations and assumptions

In scope: NIR diffuse reflectance with a portable 900 to 1700 nm spectrometer; Sri Lankan black tea; the full experimental pipeline for moisture with oven-drying reference; preprocessing, CARS, PLSR, SVR, RF and an optional neural network; held-out, source-held-out and field evaluation; a CLI or Streamlit prototype; and an evidence synthesis and framework design for polyphenols, TF:TR, caffeine, sugar and glyphosate.

Delimitations: experimental models for parameters other than moisture are built only if a partner laboratory supplies reference values; the study does not build hardware, certify results, replace laboratory testing, integrate in-line into a factory, transfer calibrations between instruments, or claim that NIR detects glyphosate at regulatory levels.

Assumptions: triplicate oven-drying gives reliable reference moisture; samples will span enough moisture variation; and a portable NIR spectrometer will be available. Findings will be limited to the teas, season and instrument tested.

1.12 Expected outcomes and deliverables

Table 4: Expected outcomes and deliverables

Deliverable	Expected output	Objective
Literature review	Thematic, parameter-by-parameter review of 40 included studies	O1
Paired dataset	NIR spectra, oven moisture, metadata and sealed subsamples	O2
Preprocessing and selection evaluation	Comparison tables and selected bands	O3, O4
Predictive models	Tuned PLSR, SVR, RF and optional neural network	O5
Model evaluation	Locked-test, source-held-out and timing results	O6
Prototype and field report	Screening tool with out-of-range warning; field accuracy and usability	O7
Feasibility map and model registry	Graded evidence and extensible design for five further parameters	O8
Research findings	Evidence-based conclusion on rapid NIR and AI tea testing	O6 to O8

1.13 Chapter summary

This chapter set out the case for rapid tea quality assessment with NIR spectroscopy and AI. Conventional reference methods are slow and parameter-specific, while existing NIR research is strong but fragmented, single-site and rarely measures time saved. The research will validate a NIR and AI framework on moisture in Sri Lankan black tea and map its feasibility for polyphenols, TF:TR, caffeine, sugar and glyphosate. Chapter 2 reviews the literature that shapes these decisions.

CHAPTER 2: LITERATURE REVIEW

2.1 Chapter overview

This chapter critically reviews research on rapid tea quality assessment with NIR spectroscopy and AI. It explains the review strategy and concept map, reviews the problem domain (the measurement task, data challenges and standards), and then reviews the technology stage by stage along the proposed pipeline: acquisition, preprocessing, wavelength selection, modelling, robustness, interpretation and training strategy. A thematic analysis compares the evidence parameter by parameter, followed by reviews of available datasets and evaluation practice. The chapter closes with a synthesis, the research positioning and the conceptual architecture.

2.2 Review strategy and concept map

2.2.1 Literature search and selection strategy

Searches covered Google Scholar, ScienceDirect, MDPI, SpringerLink, PubMed Central, RSC, IOP, Scopus and Crossref, plus Sri Lankan proceedings (iPURSE, ISAE, RESCON). Concept blocks were combined with AND and synonyms with OR, for example: (tea OR "Camellia sinensis") AND (moisture OR caffeine OR theaflavin* OR polyphenol* OR sucrose OR glyphosate) AND ("near infrared" OR NIR) AND (PLSR OR "machine learning" OR chemometric*). The date range was mainly 2016 to 2026, with seminal methods (Roberts and Smith, 1961) included.

Studies were included if they reported quantitative spectroscopic results for tea, food or agricultural quality, or defined a tea reference method; comparable matrices (feed, herbal tea, powders) were included where tea evidence was thin. Medical NIR, unrelated domains and opinion pieces were excluded. Screening ran in layers (title, abstract, full text, quality check, then backward and forward snowballing). Of 45 records (34 from the author's survey and 11 from a targeted search for caffeine, TF:TR, polyphenols, sugar and glyphosate), 5 were excluded and 40 retained. The full log is in Appendix A and the survey matrix in Appendix C.

2.2.2 Concept map

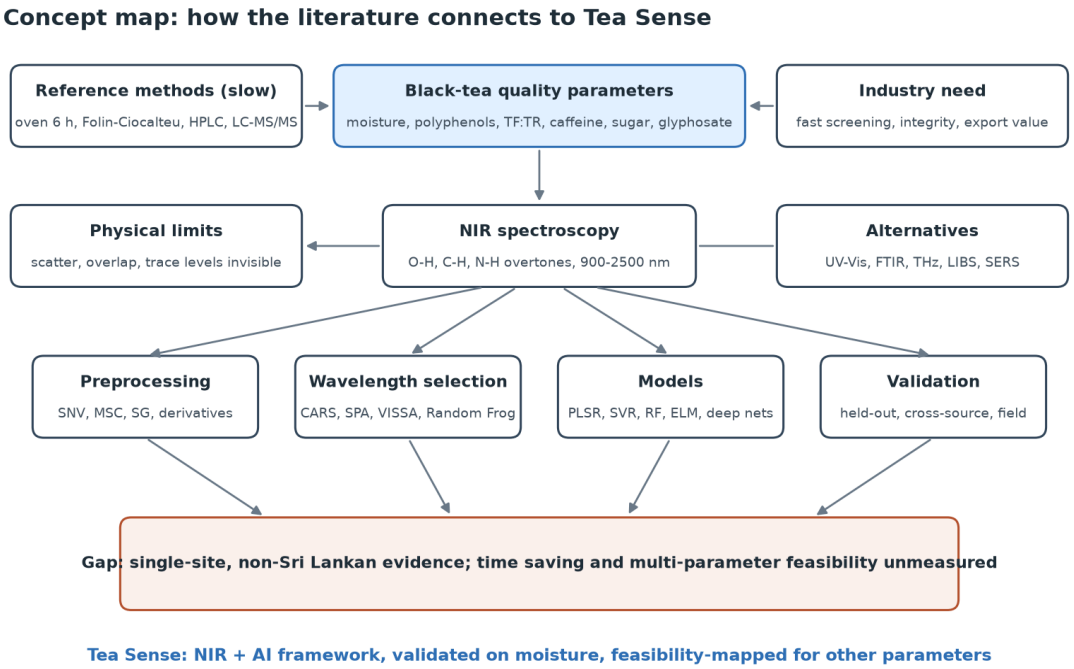


Figure 1: Concept map linking tea parameters, NIR principles, the analysis pipeline and the research gap

Figure 1 shows the logic of the review: slow reference methods and industry need define the parameters of interest; NIR reaches them through bond overtones but meets physical limits; the analysis pipeline (preprocessing, selection, models, validation) determines whether the signal becomes a reliable estimate; and the recurring weaknesses in the literature define the gap Tea Sense addresses.

2.3 Problem-domain review

2.3.1 Domain task and decision context

The target task is to estimate a continuous parameter (moisture %, and later TF:TR, caffeine or polyphenols) from one spectrum of a tea sample, or to classify a sample as adulterated. Inputs are the spectrum, sample metadata and, for training only, laboratory reference values. The decision being supported differs by user: a dryer operator decides whether to continue drying; a quality officer decides whether a lot meets specification; a broker or the Tea Board decides whether a lot needs laboratory testing for adulteration.

Errors are asymmetric. Under-predicting moisture could let tea with mould risk be stored or shipped, while over-predicting could trigger needless re-drying, which damages aroma and

loses weight. A false negative for added sugar lets adulterated tea pass. For this reason the system is positioned as screening with human review: values near specification limits or spectra outside the calibration space are flagged for laboratory confirmation, and the laboratory remains the authority (Hazarika et al., 2017; Qiu et al., 2026).

Table 5: Decisions supported and the cost of errors

User	Decision	Cost of a high prediction	Cost of a low prediction	Human review
Dryer operator	Continue or stop drying	Unneeded re-drying; aroma and weight loss	Under-dried tea stored; mould risk	Values near limit sent to laboratory
Factory QC officer	Release lot to sorting or packing	Good lot held back	Out-of-specification lot released	Flagged spectra confirmed by oven
Broker or Tea Board	Send lot for adulteration testing	Extra laboratory cost	Adulterated lot passes (sugar)	Positive screens confirmed by HPLC
Researcher	Accept or reject a model	False confidence in a weak model	Discarding a useful model	Locked test and supervisor review

2.3.2 Data and operational challenges

Narrow target range. Made tea moisture lies in a narrow band, so a model can show a low R2 even with small errors. Methodological implication: widen the calibration range with drying-stage and conditioned samples and report RMSEP and RPD alongside R2. **Sample heterogeneity.** Particle size and packing change scatter (Scheibelhofer et al., 2016). Implication: repacked triplicate scans and scatter correction. **Source and season effects.** Polyphenols vary by month and drying technique in Sri Lankan cultivars (Piyasena et al., 2026), and models often fail across varieties and regions (Qiu et al., 2026). Implication: record source metadata and test on a held-out source. **Reference error.** Oven-drying measures all volatiles at 103 C and has its own repeatability limit. Implication: triplicate references and an explicit error floor. **Small datasets.** Most tea studies have 60 to 300 samples (Andasuryani et al., 2024; Zou et al., 2022), far fewer than the number of wavelengths, causing over-fitting (Liu et al., 2024). Implication: PLSR baseline, wavelength selection and nested cross-validation. **Moisture change.** Tea exchanges moisture with air between scanning and drying. Implication: sealed bags and a short, fixed time window. **No local reference data for other parameters.** Implication: evidence synthesis and retained subsamples for later analysis.

2.3.3 Standards, regulations and professional expectations

ISO 1573 defines moisture as loss in mass at 103 C to constant mass and is the basis of the reference method. ISO 14502-1 and 14502-2 define Folin-Ciocalteu polyphenols and HPLC catechins, and the Roberts and Smith (1961) method remains the reference for TF and TR. ISO 3720 sets black-tea requirements, and importing markets apply pesticide maximum residue limits at trace levels, which is why glyphosate requires LC-MS/MS (Samanthy et al., 2021). The Sri Lanka Tea Board tests auction samples and acts against adulteration (Daily Mirror, 2018). For this project these standards mean that NIR outputs must be reported as screening estimates traceable to the standard reference methods, never as certified results, and that claims about residue safety must not be made from NIR.

2.4 Technological review by implementation stage

Each stage below compares realistic alternatives, states the selection criteria and explains the effect on later stages. The table summarises the choices.

Table 6: Summary of pipeline-stage choices

Stage	Alternatives	Criteria	Selected	Effect on later stages
Acquisition	Benchtop FT-NIR; portable InGaAs; in-line; UV-Vis, FTIR, THz, LIBS, HSI	Water sensitivity; portability; cost; tea evidence	Portable 900 to 1700 nm reflectance	Shorter range raises the importance of preprocessing
Preprocessing and QC	SNV, MSC, SG, derivatives, EMSC, wavelets	Evidence; simplicity; interpretability	Factorial comparison inside CV	Defines the space for selection
Wavelength selection	CARS, SPA, iPLS, Random Frog, VISSA, VCPA-IRIV, mRMR	Tea evidence; availability; stability	CARS plus SPA or iPLS	Fewer variables; simpler sensors
Modelling	PLSR, SVR, RF, ELM, XGBoost, ANN, CNN, BiGRU	Small-data accuracy; over-fitting risk	PLSR baseline; SVR, RF; optional NN	Needs identical splits and budgets

Stage	Alternatives	Criteria	Selected	Effect on later stages
Robustness	Random split; KS; chronological; LOSO; T2/Q flags; conformal	Honest error; safety	KS locked test, LOSO, T2/Q, bootstrap intervals	Defines reported error
Interpretation and output	Coefficients, VIP, bands; raw value; dashboard	Chemical meaning; operator clarity	VIP and bands; value, interval, flag, label	Reduces over-reliance
Training strategy	Grid, random, Bayesian, metaheuristic search	Fairness; reproducibility	Nested CV with bounded grids	Prevents tuning leakage

2.4.1 Spectral acquisition and sensing technology

Options include benchtop FT-NIR (Bruker MPA, Thermo Antaris II, Foss 5000, Buchi N-500), portable InGaAs units covering 900 to 1700 nm (NIR-S-G1, NIR-S-R2, NIR-R210, IAS-3100) and in-line analysers (Tao et al., 2026). Benchtop units cover up to 2500 nm, including the strong 1940 nm water band, but cost more and stay in the laboratory. Portable units reach the 970 and 1450 nm water bands and have predicted tea moisture, polyphenols, sugar and adulterants (Zou et al., 2022; Lan et al., 2022; Xu et al., 2023; Tang et al., 2025). Other technologies were rejected as the primary sensor: terahertz is blocked by water (Huang et al., 2024), LIBS is destructive (Zhao et al., n.d.), FTIR-ATR and UV-Vis require preparation (Damayanthi et al., 2025; 2026), and hyperspectral imaging is costly (Qiu et al., 2026). **Selection criteria:** water sensitivity, portability, cost and tea evidence. **Choice:** portable 900 to 1700 nm diffuse reflectance with three repacked scans. **Effect on later stages:** a shorter range and noisier signal make preprocessing and wavelength selection more important.

2.4.2 Preprocessing and quality control

SNV and MSC correct multiplicative scatter, SG smoothing reduces noise, and SG first and second derivatives remove baseline offsets and sharpen overlapping bands (Qiu et al., 2026). Their effect is large: preprocessing raised class separability from 53.63 to 93.20 (Das et al., 2020), and SNV with SG derivatives preceded the best CARS-PLSR model for a polyphenol ratio (Xu et al., 2026). The best combination varies by task: Zou et al. (2022) compared SNV, MSC, SG and Z-score for moisture, while Hazarika et al. (2024) compared SNV, MSC and

derivatives for CTC quality. Quality control commonly uses PCA with Hotelling T2 and Q-residuals. **Choice:** a systematic comparison of raw, SNV, MSC, SG and derivatives under fixed PLSR, fitted inside cross-validation. **Effect:** the chosen preprocessing defines the space in which wavelengths are selected.

2.4.3 Wavelength selection and feature extraction

Full spectra contain many correlated, uninformative variables. CARS iteratively keeps wavelengths with large PLS coefficients and selected few wavelengths with strong performance for CTC tea (Hazarika et al., 2024), moisture in feed (Huang et al., 2022) and a polyphenol ratio (Xu et al., 2026). Alternatives include SPA, interval PLS, Random Frog, VISSA (An et al., 2025), VCPA-IRIV (Luo et al., 2023), mRMR (Tao et al., 2026) and PCA. CARS is stochastic, so repeated runs and selection frequency are needed. **Choice:** CARS with SPA or iPLS as comparator, checked against known water bands. **Effect:** fewer wavelengths reduce over-fitting and point towards cheaper multi-band sensors.

2.4.4 Core modelling approach

PLSR is the baseline in nearly every study because it handles many collinear wavelengths with few samples. Non-linear models often help: grey-wolf-optimised SVR beat PLS for black-tea moisture (Zou et al., 2022), an optimised Elman network beat PLSR and SVR in green-tea fixation (Lan et al., 2022), ELM beat PLS for catechins (Liu et al., 2024), ELM with AdaBoost predicted TF:TR (Dong et al., 2017), and SVR beat PLSR for added sucrose (Luo et al., 2023). Deep networks performed well with thousands of spectra (Tao et al., 2026; Kang et al., 2026; Tang et al., 2025). The gains shrink at a few hundred samples.

Choice: PLSR baseline, SVR and RF as candidates, a small neural network only above about 150 samples, one model per parameter on a shared spectrum.

Effect: fair comparison needs identical splits and tuning budgets.

2.4.5 Reliability, uncertainty and robustness

Calibration results overstate performance (Revilla et al., 2024), perfect classification on modest datasets may be over-optimistic (Guo et al., 2025), and models degrade across varieties, regions and environments (Qiu et al., 2026). Robust practice includes a locked independent test set, chronological or source-based splits (Tao et al., 2026), repeat-scan repeatability, drift control with reference samples, and flagging spectra outside the calibration space (Hotelling T2 or Q-residuals).

Choice: Kennard-Stone locked test set, leave-one-source-out testing, out-of-range flags and bootstrap prediction intervals.

Effect: produces honest error estimates and safe outputs.

2.4.6 Interpretability and output presentation

NIR models are interpreted through regression coefficients, VIP scores and selected wavelengths, which can be checked against known chemistry, such as water bands near 970 and 1450 nm. Studies rarely report how results are shown to operators.

Choice: report selected bands and VIP plots for scientific interpretation, and give operators a single value with a status, a range flag and the label "screening estimate".

Effect: links model behaviour to chemistry and reduces over-reliance.

2.4.7 Training strategy and hyperparameter optimisation

Studies tune PLS latent variables by cross-validation and optimise SVR or neural networks with grid search or metaheuristics such as grey wolf (Zou et al., 2022), whale optimisation (Lan et al., 2022) or binary particle swarm (Bai et al., 2022). Metaheuristics can over-fit small datasets if tuned on the same data used for evaluation.

Choice: nested or repeated 10-fold cross-validation inside the calibration set, bounded grid search, and one-standard-error rules for latent variables; the test set is never used for tuning.

Effect: keeps the comparison fair and reproducible.

2.5 Existing work: thematic analysis

Theme 1: Moisture is the strongest and most mature NIR target

All moisture studies reviewed report usable models because water absorbs strongly in the NIR. Zou et al. (2022) modelled 305 spectra across black-tea processing stages with a 77 g spectrometer; Bai et al. (2022) reached Rp 0.971 in instant tea; An et al. (2025) and Lan et al. (2022) added machine vision; Tao et al. (2026) monitored a white-tea line with 17,856 observations; and Kang et al. (2026) covered 0 to 80% moisture with 1,505 spectra. Andasuryani et al. (2024) showed a simple PLS with oven reference on 60 herbal-tea

samples. The studies agree on feasibility but disagree on the need for non-linear models, which win mainly on large datasets.

Contradiction: high reported accuracy coexists with single-site data.

Implication: moisture is the right experimental core, with PLSR as baseline and cross-source testing.

Table 7: Comparison of NIR moisture studies

Study	Product and setting	Samples	Instrument and range	Preprocessing and selection	Best model	Validation
Zou et al. (2022)	Black tea, all processing stages, one cultivar	305 spectra	Micro-NIR 900 to 1700 nm	SNV, MSC, SG, Z-score	GWO-tuned SVR beat PLS	Split within one cultivar
Lan et al. (2022)	Green-tea fixation, one site	Series every 25 s	NIR-S-R2 900 to 1700 nm + images	SNV; fusion	WOA-Elman beat PLSR, SVR	Within one run
An et al. (2025)	Black-tea withering, controlled	Hourly series	IAS-3100 900 to 1700 nm + images	VISSA; fusion	SVR with fusion	Within one harvest
Bai et al. (2022)	Instant tea	118 samples	NIR	BPSO selection	BPSO-SVR, Rp 0.971	Calibration/validation split
Andasuryani et al. (2024)	Gambir herbal tea powder	60 (40/20)	Buchi N-500 1000 to 2500 nm	Absorbance	PLS	External 20 samples
Tao et al. (2026)	White-tea line, one season	17,856 observations	MS-580 940 to 1550 nm	MSC; mRMR	Attention BiGRU	Chronological, two test sets

Study	Product and setting	Samples	Instrument and range	Preprocessing and selection	Best model	Validation
Kang et al. (2026)	Withering, lab plus online	1,505 spectra	ATP8600, 0 to 80% moisture	Adaptive sampling	TSNet (GRU, attention, CNN)	Lab and online
Huang et al. (2022)	Cattle feed	108 (2:1)	Handheld NIR 900 to 1700 nm + THz	Derivative, SG; CARS	PLSR with fusion	Calibration/validation split

The comparison shows the pattern behind the gap: every study validates within a single product setting, and none uses Sri Lankan made tea, reports time saved against oven-drying, or holds out a whole source. Larger studies use deep models; smaller ones rely on PLSR or SVR, which matches the expected Tea Sense sample size.

Theme 2: Phenolic quality (polyphenols, catechins and TF:TR) is feasible but harder

Dong et al. (2017) predicted TF:TR, catechins and the polyphenol to amino-acid ratio with RPD above 2.5; Zareef et al. (2018) and Hazarika et al. (2024) predicted theaflavins in black tea; Zou et al. (2025) predicted TF, TR and theabrownins during rolling. For polyphenols, Xu et al. (2026) reached R2p 0.98 for a ratio, but Bai et al. (2022) obtained only Rp 0.795 for total polyphenols in instant tea from the same spectra that gave Rp 0.971 for moisture, and Liu et al. (2024) attributed difficulty to overlapping catechin bands.

Contradiction: results range from excellent to moderate depending on matrix and target.

Implication: TF:TR is the most promising extension; polyphenols need non-linear models and good reference data.

Theme 3: Caffeine and multi-parameter estimation from one spectrum

Caffeine models are consistently strong: Rehman et al. (2020) reported external RMSEP 0.08 ppm and r2 0.97 against HPLC, and Bai et al. (2022) reached Rp 0.976. Zareef et al. (2018), Chen et al. (2018) and Bai et al. (2022) all derived several parameters from one spectrum. Rehman et al. (2020) measured liquid solutions rather than leaf, limiting transfer.

Implication: a shared-spectrum, parameter-specific architecture is supported, but each parameter needs its own validated model.

Theme 4: Adulteration and safety (sugar and glyphosate)

Luo et al. (2023) quantified added sucrose in black tea with NIR (Rp 0.924, RPD 2.51), with SVR beating PLSR. Damayanthi et al. (2025; 2026) detected sugar in Sri Lankan tea with FTIR and UV-Vis, and Tang et al. (2025) detected colourants at 0.1 to 0.5 g/kg with a handheld NIR and a deep network. Piyasena et al. (2022) cautioned that natural sugars vary, complicating thresholds. Glyphosate is different: it is regulated at mg/kg, the Sri Lankan LC-MS/MS method has an LOQ of 1.8 mg/kg (Samanthy et al., 2021), and no tea study reports NIR glyphosate detection; even metals are predicted only indirectly (Revilla et al., 2024).

Implication: sugar is a feasible screening target; glyphosate needs a different rapid technology.

Theme 5: Generalisation and deployment

Most studies use random splits within one cultivar and site; few test across sources, seasons or instruments (Qiu et al., 2026; Guo et al., 2025). Portable devices are proven in the laboratory (Li et al., 2023; Tang et al., 2025), and in-line systems exist (Tao et al., 2026), but time per measurement and field accuracy are rarely reported.

Implication: Tea Sense must report cross-source and field results and measure time saved.

Table 8: Parameter feasibility matrix derived from the thematic analysis

Parameter	NIR signal	Key evidence	Best reported	Reference needed	Verdict
Moisture	Direct, strong (O-H water bands)	Zou 2022; Lan 2022; Bai 2022; Tao 2026; Kang 2026	Rp 0.971 (Bai 2022)	Oven-drying (available)	Experimental core
Caffeine	Direct (C-H, N-H), percent level	Rehman 2020; Bai 2022; Zareef 2018	Rp 0.976 (Bai 2022); r2 0.97 (Rehman 2020)	HPLC (partner)	High; extension

Parameter	NIR signal	Key evidence	Best reported	Reference needed	Verdict
TF, TR, TF:TR	Phenolic bands, percent level	Dong 2017; Zareef 2018; Zou 2025; Hazarika 2024	RPD above 2.5, R above 0.9 (Dong 2017)	Roberts and Smith (partner)	High; priority extension
Polyphenols, catechins	Overlapping phenolic bands	Bai 2022; Liu 2024; Xu 2026; Xu 2023	R2p 0.98 for TP/FAA (Xu 2026); Rp 0.795 (Bai 2022)	Folin-Ciocalteu or HPLC (partner)	Medium to high; non-linear models
Added sugar	C-H, O-H bands of sucrose	Luo 2023; Damayanthi 2025, 2026; Tang 2025	Rp 0.924, RPD 2.51 (Luo 2023)	Spiked samples plus HPLC	Medium; screening classifier first
Glyphosate	None at mg/kg	Samanthy 2021; Revilla 2024; Qiu 2026	No NIR evidence in tea	LC-MS/MS	Not feasible by NIR; SERS or strip test

2.6 Dataset review and comparison

No public dataset of Sri Lankan black-tea NIR spectra with reference values exists. Datasets in the reviewed tea studies are mostly available only on request, recorded on different instruments and ranges, and drawn from Chinese, Indian or Spanish teas. They are therefore useful for benchmarking ranges and sample sizes but cannot be merged into training. The public Tecator NIR dataset (Borggaard and Thodberg, 1992) is used only as a surrogate to test the pipeline (Section 3.13).

Table 9: Comparison of datasets in reviewed studies and the planned Tea Sense dataset

Source	Access	Samples	Structure and range	Labels	Limitations	Planned role
Zou et al. (2022)	On request	305 spectra	Micro-NIR 900 to 1700 nm	Oven moisture	One cultivar	Benchmark design

Source	Access	Samples	Structure and range	Labels	Limitations	Planned role
Lan et al. (2022)	On request	Fixation series	NIR-S-R2 900 to 1700 nm + images	Moisture	One site and run	Benchmark
Tao et al. (2026)	Not public	17,856 observations	MS-580 940 to 1550 nm	Moisture	White tea, one season	Split design
Kang et al. (2026)	Not public	1,505 spectra	ATP8600, 0 to 80% moisture	Moisture	Controlled lab	Benchmark
Bai et al. (2022)	On request	118 samples	NIR, instant tea	Moisture, caffeine, polyphenols, polysaccharides	Instant tea	Multi-parameter benchmark
Revilla et al. (2024)	Paper only	322 samples	Foss 1000 to 2498 nm	Metals, blends	Indirect metal signal	Validation lesson
Tecator (public)	Open (OpenML 505)	240 meat spectra	Infratec 850 to 1050 nm transmission	Moisture, fat, protein	Not tea	Pipeline proof of concept only
Tea Sense (planned)	Primary, controlled	Target 120 (minimum 80)	Portable 900 to 1700 nm reflectance	Oven moisture (triplicate)	One season and instrument	Training and evaluation

2.7 Evaluation and benchmarking review

Splitting protocols. Random splits dominate; Kennard-Stone or SPXY splits are used to cover spectral space; Tao et al. (2026) used a chronological split, the strongest guard against leakage. Few studies hold out a whole source. **Baselines.** PLSR is the universal baseline; mean-only baselines are rarely reported but are needed to interpret RPD. **Performance metrics.** RMSEC, RMSECV and RMSEP, R2 (or correlation R), and RPD are standard; RPD

above 2.5 is generally treated as good for quantification and above 3 as excellent. Bias is seldom reported despite mattering for sale weight. **Reliability.** Repeat-scan standard deviation and calibration-to-validation gaps are the main reliability measures. **Robustness.** Cross-source, cross-season and field testing is rare (Qiu et al., 2026). **Efficiency.** Measurement time is claimed as "rapid" but rarely quantified; even Rehman et al. (2020) report only the HPLC run time. **Statistical uncertainty.** Most studies report single-split point estimates without confidence intervals or significance tests.

Table 10: Common interpretation of RPD in NIR studies

RPD	Interpretation	Use for Tea Sense
Below 1.5	Not usable	Model rejected
1.5 to 2.0	Distinguishes high from low values only	Coarse screening at best
2.0 to 2.5	Approximate quantification	Minimum acceptable
2.5 to 3.0	Good quantification	Target
Above 3.0	Excellent	Stretch goal

Implication: Tea Sense will report RMSEC, RMSECV, RMSEP, R2p, RPD and bias against a mean-only and a PLSR baseline, use a locked Kennard-Stone test set plus leave-one-source-out testing, apply paired Wilcoxon tests and bootstrap confidence intervals, and measure time per sample directly.

2.8 Synthesis and final research positioning

Established: NIR reliably estimates moisture and, with more effort, caffeine, TF:TR and polyphenols; preprocessing and wavelength selection matter; non-linear models can beat PLSR, especially with more data; portable devices work.

Unresolved: performance on Sri Lankan black tea; generalisation across sources; time saved and field accuracy; and a consolidated view of which parameters one framework can cover.

Why these methods: oven-drying is the accepted reference and is available; PLSR, SVR and RF match the expected sample size; CARS is proven for tea; locked and source-held-out tests answer the generalisation gap.

Table 11: Positioning against the closest studies

Study	What it did	How Tea Sense differs
Zou et al. (2022)	Black-tea moisture, micro-NIR, one cultivar	Sri Lankan tea, several sources, cross-source and field tests, time measured
Bai et al. (2022)	Four parameters from one spectrum, instant tea	Leaf black tea; graded feasibility including sugar and glyphosate
Dong et al. (2017)	TF:TR and catechins during fermentation	Treats TF:TR as a priority extension in a shared framework
Tao et al. (2026)	In-line deep model, white tea, one season	Portable, low-cost, small-data methods with honest baselines
Luo et al. (2023)	Added sucrose in black tea	Places sugar screening within a multi-parameter roadmap

2.9 Proposed conceptual architecture

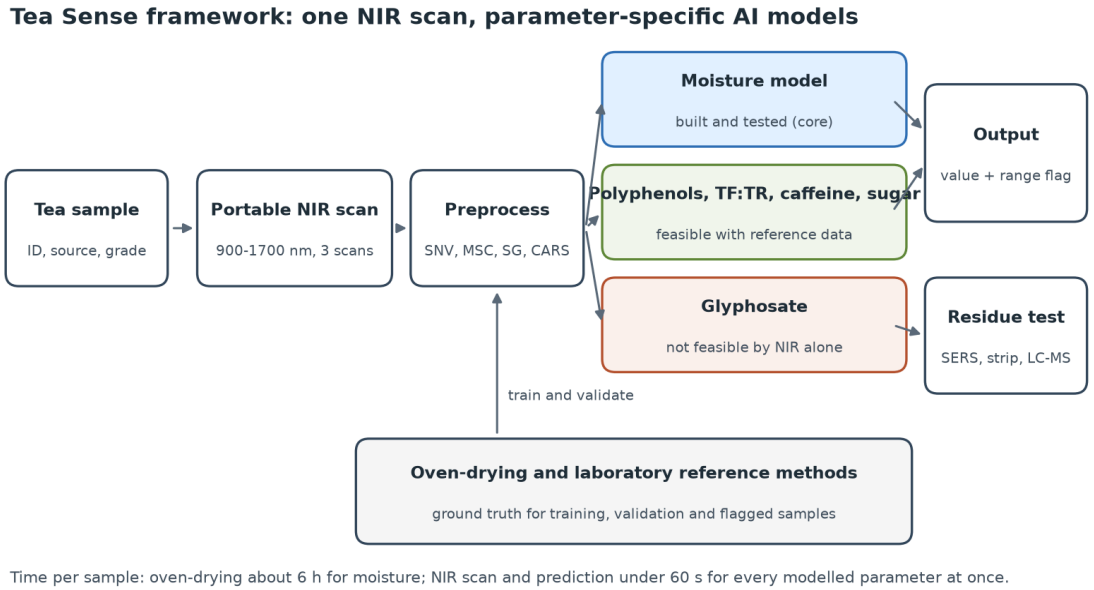


Figure 2: Tea Sense conceptual architecture

Figure 2 shows the shared-spectrum design that follows from the review. A tea sample is scanned once with a portable NIR spectrometer, preprocessed once, and passed to a registry of parameter-specific models. The moisture model is built and tested in this project.

Polyphenols, TF:TR, caffeine and sugar models are added when reference data from a partner laboratory are available, as the evidence in Section 2.5 supports their feasibility. Glyphosate is routed to a separate rapid residue test, because NIR cannot see it. Oven-drying and laboratory methods remain the ground truth for training, validation and flagged samples. Each output carries a value, a calibration-range flag and a screening label. The detailed architecture and data flow are in Appendix I.

2.10 Chapter summary

The review shows that NIR with AI is well proven for moisture, promising for caffeine, TF:TR and polyphenols, emerging for added sugar and unsuitable for trace glyphosate. Preprocessing, wavelength selection, model choice and validation design all strongly affect results, and most evidence is single-site, non-Sri Lankan and silent on time saved. Tea Sense is therefore positioned as a shared-spectrum framework validated on moisture with honest generalisation testing and a graded feasibility map for other parameters.

CHAPTER 3: RESEARCH, TECHNICAL AND PROJECT METHODOLOGY

3.1 Chapter overview

This chapter explains how the research questions will be answered. It sets out the research philosophy using Saunders' Research Onion, the experimental design and hypotheses, and the technical pipeline stage by stage. It then covers requirements engineering, system design, data management, development methodology and the evaluation methodology, followed by project management, resources, risks and a preliminary proof of concept.

3.2 Research methodology using Saunders' Research Onion

Table 12: Research Onion justification

Layer	Choice	Justification against the research questions
Philosophy	Positivism (pragmatism for the prototype)	RQ1 to RQ4 concern objectively measurable error against a reference method
Approach	Deductive	Hypotheses H1 to H5 are derived from the literature and tested
Methodological choice	Quantitative, supported by structured evidence synthesis	Spectra and reference values are numerical; RQ5 parameters lack in-house reference data, so evidence is graded
Strategy	Experiment (moisture) and systematic review (other parameters)	Controlled comparison of preprocessing, selection and models; evidence grading for RQ5
Time horizon	Cross-sectional	One collection season within one academic year; seasonal effects stated as a limitation
Techniques	Gravimetric reference, NIR acquisition, statistical modelling, evidence matrix, SUS questionnaire	Standard NIR practice; SUS supports RQ4 usability

3.3 Research design and hypotheses

Independent variables: preprocessing (raw, SNV, MSC, SG, derivatives), wavelength set (full, CARS, SPA/iPLS), model (PLSR, SVR, RF, optional neural network), and measurement setting (laboratory, field).

Dependent variables: RMSEP, R2p, RPD, bias, time per measurement, repeat-scan SD and SUS.

Baselines: mean-only and PLSR on raw spectra.

Ablations: remove preprocessing, selection or the non-linear model from the best pipeline.

Confounders and controls: sample source (stratified split and LOSO), scanning order (randomised), instrument drift (white and dark references, control sample), time between scan and drying (fixed window), and operator (single trained operator).

Table 13: Hypotheses

ID	Hypothesis	RQ
H1	NIR moisture models reach R2p of at least 0.90 and RPD of at least 2.5 on held-out Sri Lankan black tea.	RQ1
H2	Scatter-corrected and smoothed spectra give lower RMSECV than raw spectra; CARS matches the full spectrum with fewer variables.	RQ2
H3	At least one machine-learning model beats PLSR on the locked test set (paired Wilcoxon, $p < 0.05$), and LOSO RMSEP stays within 1.5 times the test RMSEP.	RQ3
H4	NIR reduces measurement time from about 6 h to under 60 s per sample, with field RMSEP within 1.5 times laboratory RMSEP.	RQ4
H5	Evidence supports NIR feasibility for caffeine, TF:TR and polyphenols, partial feasibility for sugar, and not for glyphosate.	RQ5

3.4 Proposed technical methodology and implementation pipeline

3.4.1 Sample acquisition, data ingestion and governance

Aspect	Detail
Inputs	Black-tea samples from several Sri Lankan sources and grades; drying-stage and humidity-conditioned samples
Processing	Assign coded IDs; log source, grade, stage, date, temperature and humidity; seal; scan three times with repacking after white and dark references; oven-dry 3 to 5 g in triplicate at 103 C to constant mass; seal a retained subsample
Alternatives	Retail-only samples; single-scan protocol; halogen analyser reference
Selected	Mixed sources with triplicate scans and triplicate oven reference
Outputs	Raw spectra files, mass log, metadata form, retained subsamples
Testing	Pilot of 10 to 15 samples checks repeatability and drying time
Fallback	Retail teas of several grades if suppliers withdraw

3.4.2 Data quality assessment

Aspect	Detail
Inputs	Raw spectra and reference triplicates
Processing	Check replicate SD (at most 0.2%); PCA with Hotelling T2 and Q-residuals; control-sample drift check; moisture histogram
Alternatives	Visual inspection only; robust PCA
Selected	Rule-based QC with logged exclusions
Outputs	Clean dataset v2 and QC report
Testing	Re-run QC script on raw data reproduces the same exclusions
Fallback	Re-scan or re-dry flagged samples

3.4.3 Pre-processing

Aspect	Detail
Inputs	Averaged spectra (calibration set only for fitting)

Aspect	Detail
Processing	Raw, SNV, MSC, SG smoothing (window 7 to 21, order 2), SG first and second derivatives, and combinations, inside cross-validation
Alternatives	EMSC, detrending, wavelet denoising
Selected	Factorial grid of the standard methods
Outputs	RMSECV table per preprocessing
Testing	Fixed PLSR and folds for fair comparison
Fallback	Raw mean-centred spectra if no method helps

3.4.4 Baseline construction

Aspect	Detail
Inputs	Calibration set
Processing	Mean-only predictor; PLSR on raw mean-centred spectra with latent variables chosen by RMSECV (one-standard-error rule)
Alternatives	MLR on selected bands
Selected	Mean-only (floor) and PLSR (strong chemometric baseline)
Outputs	Baseline RMSEP, R2p, RPD
Testing	Same locked test set as all candidates
Fallback	Not applicable

3.4.5 Proposed model and system

Aspect	Detail
Inputs	Preprocessed spectra (full or selected bands)
Processing	CARS (repeated runs) and SPA/iPLS; PLSR, SVR (RBF; C, gamma, epsilon), RF (trees, depth, features); optional small neural network above about 150 samples; one model per parameter in a registry sharing preprocessing
Alternatives	XGBoost, ELM, 1D-CNN, BiGRU
Selected	PLSR, SVR, RF, optional neural network
Outputs	Tuned, saved pipelines

Aspect	Detail
Testing	Nested CV; locked test; paired Wilcoxon against PLSR
Fallback	PLSR alone if non-linear models do not beat it

3.4.6 Reliability, uncertainty and robustness component

Aspect	Detail
Inputs	Trained models and new spectra
Processing	Hotelling T2 and Q-residual limits from calibration data; bootstrap prediction intervals; LOSO testing; repeat-scan SD
Alternatives	Conformal prediction; Gaussian process regression
Selected	T2/Q flags with bootstrap intervals
Outputs	Prediction with interval and out-of-range flag
Testing	Flag rate on test and field data; interval coverage
Fallback	Range flag only

3.4.7 Output, explanation and presentation component

Aspect	Detail
Inputs	Prediction, interval, flag, selected bands
Processing	CLI and Streamlit prototype showing value, interval, status and "screening estimate" label; VIP and band plots for analysts
Alternatives	Mobile app; notebook only
Selected	CLI first, Streamlit if time allows
Outputs	Prototype and reports
Testing	Field trial with SUS questionnaire
Fallback	CLI only

3.4.8 Reproducibility and experiment tracking

Aspect	Detail
Inputs	Code, configuration, data versions
Processing	Git for code; MLflow or CSV logs for runs; fixed seeds; pinned library versions; read-only raw data; hashed test set
Alternatives	Weights and Biases; DVC
Selected	Git with MLflow or CSV logs
Outputs	Traceable results
Testing	Clean re-run reproduces reported metrics
Fallback	Structured CSV logging only

3.5 Requirements engineering

3.5.1 Stakeholder analysis

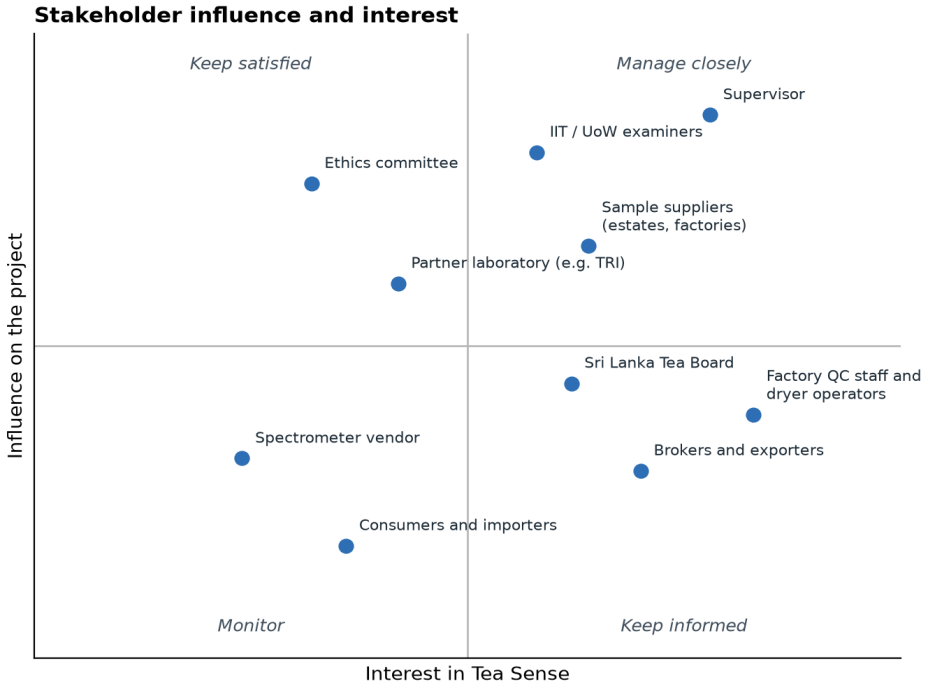


Figure 3: Stakeholder influence and interest

The supervisor, examiners and sample suppliers have high influence and high interest and are managed closely: supplier withdrawal is a key data risk. The ethics committee and a partner laboratory have influence over what can be done but less day-to-day interest and are kept

satisfied through early approval and clear requests. Factory QC staff, dryer operators, brokers and the Tea Board are the eventual beneficiaries; their main risk is over-reliance on predictions, so they are kept informed and involved in the field test. The full stakeholder onion is in Appendix F.

3.5.2 Requirement elicitation and evidence

Requirements were derived from four sources: the literature (Chapter 2), standards and reference methods (ISO 1573, ISO 14502, Roberts and Smith), industry context (Tea Board actions and export data), and the pilot study. Each requirement traces to a gap, objective or risk. Short informal discussions with a tea technologist and field-test users are planned to validate usability requirements; these involve human participants and will take place only after ethics approval (Appendix N). Instruments are in Appendix F.

3.5.3 Functional and non-functional requirements

Table 14: Key requirements (MoSCoW); full specification in Appendix G

ID	Priority	Requirement	Verification method	Traceability
FR01	Must	Acquire referenced spectra, three scans per sample	Inspection of raw files	O2, G1
FR02	Must	Record triplicate oven moisture and test time	Mass log audit	O2, O6
FR03	Must	Configurable preprocessing inside CV	Unit test; leakage test	O3, G2
FR04	Must	Lock and hash test set before modelling	Hash check	O6, G3
FR05	Must	CARS plus comparator selection	Experiment log	O4, G2
FR06	Must	Train PLSR, SVR, RF on identical splits	Run log	O5, G3
FR07	Must	Model registry, one model per parameter	Code review; add mock parameter	O8, G5
FR08	Must	Predict with out-of-range flag	Test with spiked out-of-range spectra	O7
FR09	Should	Retain sealed subsamples	Sample log	O8

ID	Priority	Requirement	Verification method	Traceability
FR10	Could	Neural model if about 150+ samples	Run log	O5
NFR01	Must	R2p at least 0.90, RPD at least 2.5	Locked-test evaluation	H1
NFR02	Must	Under 60 s per sample	Timed trials	H4
NFR03	Must	Fully reproducible	Clean re-run	Integrity
NFR04	Must	Screening label; no personal data	UI review; data audit	Chapter 4
NFR05	Should	SUS at least 68	Questionnaire	O7

3.6 System and software design

The system has five components: an acquisition module (spectrometer export and metadata form), a data store (read-only raw files, versioned clean dataset), an analysis pipeline (QC, preprocessing, selection, training, evaluation, built on scikit-learn pipelines so that fitting happens only on calibration data), a model registry (one saved pipeline per parameter with its validation record and calibration limits), and a prototype (CLI, then Streamlit) that loads a new spectrum, applies the registry and returns values, intervals and flags. The main use cases are register sample, calibrate spectrometer, scan, record oven reference, train and evaluate, predict, review flagged result and export report. The context and use-case diagrams are in Appendix H and the component and data-flow diagrams in Appendix I.

3.7 Data requirements and management plan

Table 15: Dataset plan

Item	Plan
Role	Primary paired dataset for training and evaluation; Tecator only for pipeline testing
Schema	sample_id, source_code, grade, stage, date, temp_C, RH_pct, scan_1..3 (absorbance per wavelength), moisture_rep_1..3, moisture_mean, oven_minutes, retained_subsample
Split strategy	About 70:30 Kennard-Stone calibration and locked test; leave-one-source-out where at least three sources exist

Item	Plan
Sample-level separation	All scans and replicates of one sample stay in the same set
Storage and backup	Raw data read-only; two cloud backups plus local copy
Versioning	Data versions hashed; derived data produced by scripts; Git for code
Retention	Data kept until degree award plus one year, then anonymised dataset archived
Licensing and ethics	Supplier consent; coded sources; no personal data; Tecator used under its stated terms with instrument acknowledged

3.8 Development methodology

An iterative CRISP-DM-style approach was chosen over Waterfall, which assumes fixed requirements, and full Scrum, which adds ceremony unsuited to a single developer. Each two-week iteration adds one pipeline component (for example, a preprocessing method), evaluates it by cross-validation and records the result. **Definition of done:** code committed, unit and leakage tests pass, results logged, and the supervisor update written. **Testing:** unit tests for preprocessing functions, a leakage test confirming the test set is untouched, and an end-to-end re-run. **Change control:** any change to the protocol or scope is agreed with the supervisor and logged; changes affecting participants or data are cleared with ethics first.

3.9 Evaluation methodology

3.9.1 Dataset splits and validation protocol

The test set (about 30%) is selected by Kennard-Stone, checked to span the moisture range, hashed and locked before any modelling. All preprocessing, wavelength selection and tuning are fitted inside nested or repeated 10-fold cross-validation on the calibration set using scikit-learn pipelines, and the test set is used once per shortlisted pipeline. Replicate scans of a sample never cross sets. Where at least three sources exist, leave-one-source-out validation measures generalisation. No tuning ever uses test or field data.

3.9.2 Baselines and ablation studies

Simple baseline: mean-only predictor.

Strong baseline: PLSR on raw mean-centred spectra.

Ablations: from the best pipeline, remove (a) preprocessing, (b) wavelength selection and (c) the non-linear model, each under the same folds and tuning budget, to show what each component contributes.

3.9.3 Metrics and analysis

Table 16: Metrics defined before experimentation

Metric	What it tests
RMSEP	Typical error in % moisture on unseen samples (primary metric)
R2p and RPD	Variance explained; RPD = SD of reference / RMSEP guards against narrow ranges
Bias	Systematic over- or under-prediction affecting sale weight
RMSEC vs RMSECV vs RMSEP	Over-fitting
Paired Wilcoxon test; bootstrap 95% CI	Whether differences between models are real
LOSO RMSEP	Generalisation to an unseen source
Repeat-scan SD	Measurement repeatability
Seconds per sample	Time saved versus oven-drying
Interval coverage; flag rate	Reliability of uncertainty outputs
SUS score	Usability in the field

3.9.4 Research question to evaluation traceability

Table 17: Research question to evaluation traceability

RQ	Experiment	Evidence	Decision supported
RQ1	Locked-test evaluation of the best pipeline	RMSEP, R2p, RPD, bias	Is NIR accurate enough for moisture screening?
RQ2	Preprocessing grid; CARS vs SPA/iPLS vs full	RMSECV tables, band stability	Which preprocessing and bands to deploy
RQ3	PLSR vs SVR vs RF (vs NN); LOSO	RMSEP, paired tests, LOSO error	Whether AI models are worth their complexity
RQ4	Timed oven and NIR tests; field trial	Seconds per sample, field RMSEP, SUS	Whether NIR is practical outside the lab

RQ	Experiment	Evidence	Decision supported
RQ5	Evidence grading; optional partner-lab pilot	Feasibility matrix	Which parameters to add next

3.10 Project management methodology

The project is managed as ten work packages with milestones, a prioritised backlog (MoSCoW, with Must items scheduled first), fortnightly supervisor meetings with written notes, a weekly progress review against the Gantt chart, and an issue log (Appendix K). Contingency is built in through overlapping phases and droppable Could items.

3.10.1 Schedule and milestones

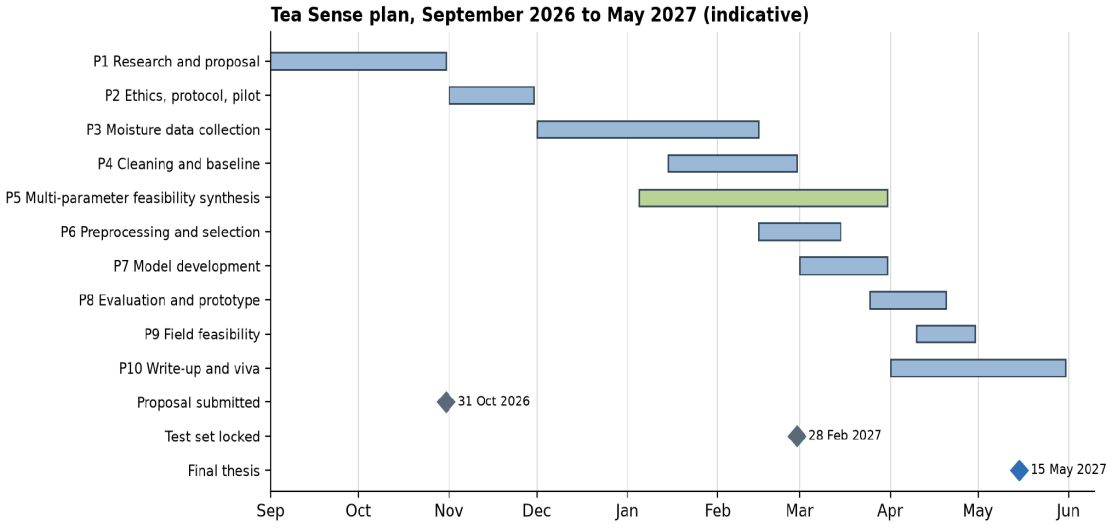


Figure 4: Summary schedule and milestones

Data collection (December to February) is the critical path; baseline modelling starts on early batches, and the multi-parameter synthesis runs in parallel. Two to three weeks of contingency sit within P10. The task-level chart and backlog are in Appendix J.

3.10.2 Deliverables and dates

Table 18: Deliverables and dates

Deliverable	Objective	Due date	Acceptance criterion	Dependency
Project proposal	O1	31 Oct 2026	Approved by supervisor	Literature review

Deliverable	Objective	Due date	Acceptance criterion	Dependency
Ethics approval and SOPs	O2	30 Nov 2026	Signed approval; pilot done	Proposal
Paired dataset	O2	15 Feb 2027	At least 80 samples; SD at most 0.2%	Spectrometer; suppliers
Locked test set and baseline	O5, O6	28 Feb 2027	Hash recorded; baseline logged	Dataset
Feasibility matrix	O8	31 Mar 2027	All six parameters rated	Review
Model comparison	O3 to O5	31 Mar 2027	All runs logged	Baseline
Evaluation and prototype	O6, O7	20 Apr 2027	Metrics reported; prototype runs	Models
Field report	O7	30 Apr 2027	Field RMSEP and SUS reported	Prototype
Final thesis	All	15 May 2027	Submitted	All

3.11 Resources and new technical skills

Hardware: portable NIR spectrometer (model and access to confirm), laboratory oven, analytical balance (0.001 g), desiccator, sealed bags, temperature and humidity logger, laptop. **Software:** Python, scikit-learn, SciPy, pandas, matplotlib, optional PyTorch, Git, MLflow or CSV logs, Streamlit. **Compute and storage:** a laptop CPU is sufficient; two cloud backups. **Domain expertise:** supervisor, and a tea technologist or partner laboratory if available. **New skills:** NIR chemometrics (CARS, PLSR diagnostics), laboratory moisture protocol and spectrometer operation, learned through the reviewed literature, online chemometrics courses, the pilot study and supervisor guidance. **Constraints:** no in-house HPLC, spectrophotometer or LC-MS/MS.

3.12 Risk analysis and mitigation

Table 19: Highest-priority risks; full register in Appendix K

Risk	L	I	Trigger	Prevention	Contingency	Owner
Spectrometer unavailable	M	H	No confirmed unit by mid-November	Confirm early; second source	Benchtop at partner lab	Student
Too few samples or narrow moisture range	M	H	Under 50 samples by January; SD of moisture under 1%	Several suppliers; conditioning	Accept 80; drop neural model	Student
Inconsistent oven reference	M	H	Replicate SD above 0.2%	SOP; pilot; triplicates	Re-run; exclude with reason	Student
Data leakage	M	H	Test error far below CV error	Locked hash; pipelines	Re-split; disclose	Student
No partner lab for other parameters	H	M	No agreement by January	Early outreach; retained subsamples	Evidence-only assessment (planned)	Student, supervisor
Scope creep to all parameters	H	H	New parameter added to experiments	MoSCoW; moisture-first agreed	Move to future work	Student, supervisor
Field conditions degrade accuracy	H	M	Field RMSEP above 1.5 times lab	Log conditions; warm-up	Report as feasibility finding	Student

3.13 Preliminary proof of concept and feasibility evidence

Preliminary; surrogate data, not tea. To test the planned pipeline before tea data exist, it was run on the public Tecator dataset: 240 meat samples measured on a Tecator Infratec Food and Feed Analyzer (850 to 1050 nm transmission) with reference moisture from 32.8 to 76.6% (Borggaard and Thodberg, 1992). A Kennard-Stone split gave 168 calibration and 72 locked test samples, and all preprocessing and tuning were fitted with 10-fold cross-validation on the calibration set only.

Table 20: Preliminary proof of concept on surrogate NIR data (held-out test set)

Model	RMSEC (%)	RMSEP (%)	R2p	RPD	Bias (%)
Mean-only baseline	11.332	10.364	-0.088	0.97	-2.951
PLSR (SNV + SG 1st derivative)	1.704	1.921	0.963	5.21	-0.263
CARS-PLSR (11 of 100 wavelengths)	1.643	1.959	0.961	5.11	-0.096
SVR	0.685	1.044	0.989	9.58	0.058
Random Forest	0.479	1.298	0.983	7.71	0.043

Scatter correction mattered: SNV lowered PLSR RMSECV from 2.761% (raw) to 1.955%. CARS kept 11 of 100 wavelengths with almost unchanged test error (1.959% against 1.921%). SVR beat PLSR (RMSEP 1.044% against 1.921%; paired Wilcoxon $p < 0.001$), and a prediction took about 1.5 ms of computation. These results confirm the code and validation protocol work end to end; they do not predict tea performance, because the matrix, range and instrument differ. Plots and logs are in Appendix M.

3.14 Chapter summary

The research is a positivist, deductive, quantitative experiment for moisture, supported by structured evidence synthesis for other parameters. An eight-stage pipeline, fixed baselines, ablations, a locked test set and source-held-out validation will answer RQ1 to RQ4, and a graded feasibility matrix will answer RQ5. Requirements, design, data management, schedule, resources and risks have been defined, and a preliminary run on surrogate data confirms that the pipeline functions.

CHAPTER 4: LEGAL, ETHICAL, SOCIAL AND PROFESSIONAL ISSUES

4.1 Chapter overview

This chapter identifies the social, legal, ethical and professional issues specific to Tea Sense, mapped to the BCS Code of Conduct (Public Interest; Professional Competence and Integrity; Duty to Relevant Authority; Duty to the Profession), and the responsible use of generative AI. The mitigations are already embedded in the methodology in Chapter 3.

4.2 Social issues

Risk	Stakeholder	Consequence	Control	Contingency	Evidence
Unequal performance across sources	Smaller or under-sampled estates	Wrong moisture or quality values affect their sale weight	Diverse sourcing; per-source RMSEP (Public Interest)	State validated scope; collect more data	Per-source residual table
Over-reliance and automation bias	Factory staff, brokers	Mouldy or poor tea released on a model alone	Screening label, interval, range flag; lab confirmation	Withdraw prototype from use	UI review; field interviews
Accessibility	Operators with limited English or technical training	Misread outputs	Large numerals, text plus colour, few steps	Sinhala and Tamil labels as future work	SUS and observed errors
Exclusion by cost	Small producers	Benefits limited to large factories	Low-cost sensor and free software	Document a minimal set-up	Cost table in thesis

4.3 Legal issues

Data protection and consent. The dataset holds no personal data. Field participants give written consent; responses are anonymous, and consent forms are stored securely under the Sri Lanka Personal Data Protection Act No. 9 of 2022 and university policy. **Commercial confidentiality.** Supplier identities are coded and named only with written consent. **Licences and copyright.** Python libraries are used under BSD, MIT or Apache licences; vendor software under its licence; the Tecator data under its stated terms, acknowledging the instrument and company; all literature is cited. **Third-party tools and storage.** No confidential data are entered into third-party AI tools; cloud backups use university-approved storage, avoiding unnecessary cross-border transfer. **Sector rules.** No claim of compliance with ISO 3720, residue limits or buyer specifications is made.

4.4 Ethical issues

Ethics approval will be obtained before data collection or field testing (Appendix N). **Data minimisation:** only metadata needed for analysis is collected. **Bias and fairness:** per-source reporting and LOSO testing expose unequal performance. **Transparency and error communication:** every output shows an interval, a flag and a screening label, and published results report calibration and prediction errors side by side. **Human oversight:** laboratory confirmation remains the authority. **Non-fabrication:** no data, results or feedback are invented, negative results are reported, and glyphosate is explicitly not claimed as detectable by NIR.

4.5 Professional issues

Work stays within competence, with supervisor and domain review of the protocol (Professional Competence and Integrity). Reporting is accurate and reproducible through locked tests, logs and clean re-runs (Duty to Relevant Authority). Code is developed with version control, tests and dependency pinning, and no credentials or supplier data are committed. Any flaw found in the method or published code will be disclosed and corrected. The work follows the BCS Code of Conduct and cites all sources (Duty to the Profession).

4.6 Responsible use of Generative AI tools

4.6.1 Intended and permitted use

Claude, Gemini was used for language refinement, structuring the proposal against the template, drafting assistance for sections that the author then reviewed and edited, literature-search support, and help writing preliminary analysis code that was run and inspected by the author.

4.6.2 Prohibited or high-risk use

Generative AI must not be used to fabricate citations, data or results, to generate experimental findings, to process confidential supplier data, to disguise plagiarism, or to make unverified domain claims. Any AI-suggested source that cannot be located and read is removed.

4.6.3 Verification and human accountability

Every factual claim is verified against the original source, every cited paper is read by the author before submission, AI-assisted code is run and inspected, and evidence (search records, logs, drafts) is retained. Items not yet verified are marked in the references. The author is accountable for all content.

4.6.4 Privacy, confidentiality and intellectual property

Supplier names, sample data, personal data and unpublished results are never entered into AI tools; minimal synthetic or public examples (such as the Tecator data) are used instead. Copyrighted text is not pasted for rewriting.

4.6.5 Transparency, disclosure and AI-use register

All uses are disclosed in the front-matter declaration and recorded in the AI-use register in Appendix L, which lists the tool, date, purpose, input category, verification performed and whether the output was retained, rewritten or rejected.

4.6.6 Generative AI declaration

Generative AI tools were used to support language refinement, structuring, drafting assistance, literature-search support and preliminary code. All technical decisions, objectives, methods and interpretations were reviewed and approved by the author, who takes full responsibility for the accuracy, originality and integrity of this proposal. This statement is consistent with the declaration in the front matter.

4.6.7 Project-specific Generative AI risk assessment

Risk	Possible consequence	Control and evidence
Fabricated or misattributed citation	Academic misconduct; false evidence	Each DOI checked via Crossref or publisher; markers; reading log
Incorrect technical claim	Flawed method or examiner challenge	Claims checked against papers; supervisor review
Over-reliance on AI drafting	Student cannot defend the work	Author rewrites sections; viva rehearsal
Confidential data exposure	Breach of supplier trust	No supplier data entered; register audit
Unverified code	Wrong results	Code run, inspected and unit-tested; logs kept

4.7 Chapter summary

The main risks are over-reliance on screening outputs, unequal performance across sources, supplier confidentiality, misrepresentation of results and misuse of generative AI. Each has a preventive control, a contingency and evidence that the control is followed, and these are built into the methodology. These measures reduce, but do not eliminate, the risks, so they will be reviewed throughout the project.

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Zareef, M., Chen, Q., Ouyang, Q., Kutsanedzie, F.Y.H., Hassan, M.M., Viswadevarayalu, A. et al. (2018) Prediction of amino acids, caffeine, theaflavins and water extract in black tea using FT-NIR spectroscopy coupled chemometrics algorithms. Analytical Methods, 10(25), pp.3023-3031. https://doi.org/10.1039/C8AY00731D

APPENDICES

Each appendix supports a section of the main text; the essential arguments remain in Chapters 1 to 4.

Appendix A: Detailed search strategy and screening log

Search	Databases	String (abridged)	Date	Records	Kept
S1 NIR tea quality	Google Scholar, ScienceDirect, MDPI	(tea) AND ("near infrared" OR NIR) AND (quality OR chemometric*)	Before Sep 2026 (author survey)	14	11
S2 Tea moisture	Google Scholar, MDPI, SpringerLink, Nature, IOP	(tea) AND (moisture OR "water content") AND (NIR OR spectroscop*)	Before Sep 2026	9	9
S3 Sri Lankan tea chemistry	Proceedings, BNRC, Applied Food Research	("Sri Lanka*" OR Ceylon) AND tea AND (sugar OR polyphenol*)	Before Sep 2026	6	6
S4 Off-domain NIR captured by broad searches	Various	"near infrared spectroscopy" (broad)	Before Sep 2026	5	0
S5 Targeted: caffeine, TF:TR, catechins	PMC, RSC, SpringerLink, Crossref	(tea) AND (caffeine OR theaflavin* OR catechin*) AND NIR	19 Oct 2026	7	7
S6 Targeted: sugar, glyphosate	PMC, Peradeniya repository, Crossref	(tea) AND (sucrose OR glyphosate) AND (NIR OR LC-MS)	19 Oct 2026	4	4

Screening: 45 records; 5 excluded at full text as off-domain (gemology, two cerebral-oximetry papers, a neuroscience commentary, an RFID antenna paper); 40 included. Counts per search in S1 to S3 are reconstructed from the author's survey sheet and should be confirmed against the original search history.

Appendix B: Concept map

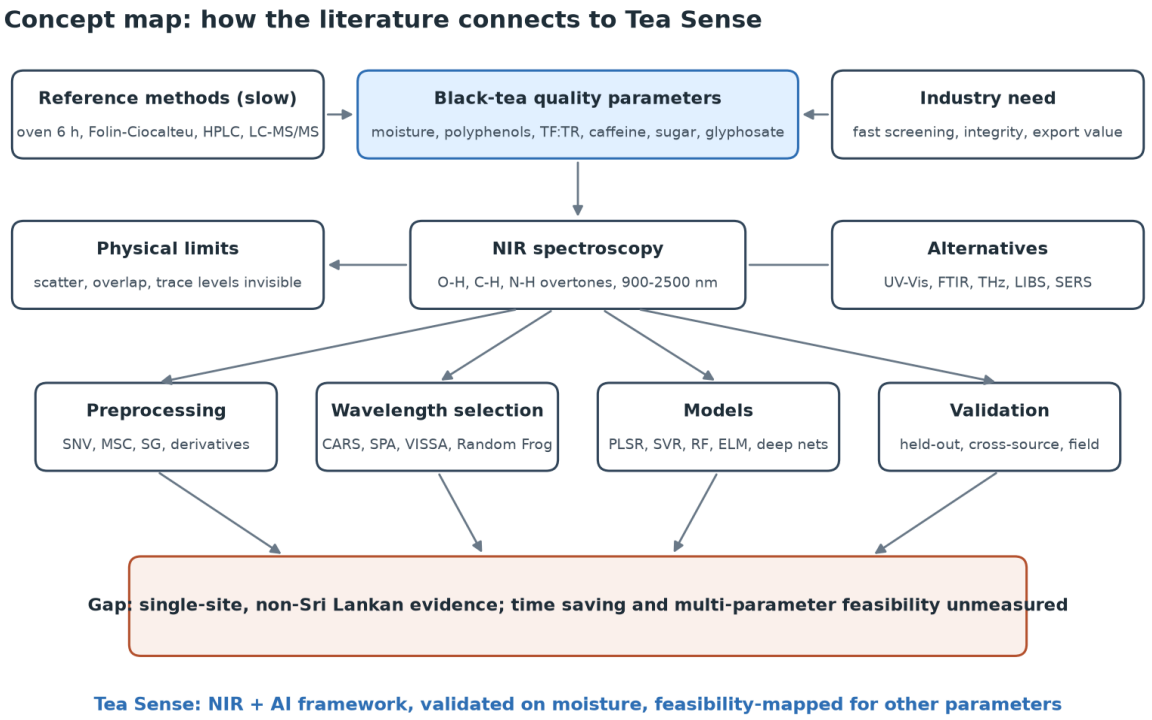


Figure 5: Concept map (full size)

Appendix C: Literature survey matrix

The full matrix of all 45 records (34 from the author's survey and 11 marked NEW) is presented on the landscape pages that follow the appendices.

Appendix D: Research gap analysis matrix

Gap	Moisture	Polyphenols	TF:TR	Caffeine	Sugar	Glyphosate
NIR evidence exists	Strong	Moderate to strong	Strong	Strong	Emerging	None
Sri Lankan black-tea evidence	None	Chemistry only (Piyasena 2026)	None	None	FTIR, UV-Vis only	LC-MS only
Cross-source validation	Rare	Rare	Rare	Rare	Rare	Not applicable
Time saving measured	Rare	No	No	HPLC time only	No	No
Addressed by Tea Sense	Experimentally	Evidence plus optional pilot	Evidence plus optional pilot	Evidence plus optional pilot	Evidence plus optional pilot	Evidence (negative)

Appendix E: Dataset comparison matrix and data dictionary

The dataset comparison is in Section 2.6. The Tea Sense data dictionary is below.

Field	Type	Unit	Description
sample_id	string	-	Coded ID, e.g. S1-017
source_code	string	-	Coded estate or factory
grade	category	-	BOP, BOPF, Pekoe, Dust, etc.
stage	category	-	Made tea, final dryer, conditioned
date	date	-	Scan date
temp_C, RH_pct	float	C, %	Ambient conditions at scan
scan_k_w	float	absorbance	Scan k at wavelength w
moisture_rep_1..3	float	% wet basis	Oven replicates
moisture_mean	float	% wet basis	Reference value
oven_minutes	int	min	Time to constant mass
retained_subsample	bool	-	Sealed subsample kept for partner-lab analysis
qc_flag	string	-	QC exclusion reason, if any

Appendix F: Stakeholder onion, elicitation instruments and evidence

Layer	Stakeholders
Core (system)	Researcher; prototype
Operational	Factory QC staff, dryer operators, laboratory analysts
Containing business	Supervisor, examiners, sample suppliers, partner laboratory
Wider environment	Sri Lanka Tea Board, brokers, exporters, importers, consumers, ethics committee, spectrometer vendor

Planned instruments (after ethics approval): (1) a short discussion guide for a tea technologist: how moisture is currently checked, acceptable error, decision points, sample handling; (2) a field-test observation sheet: time per scan, handling errors, conditions; (3) the 10-item System Usability Scale.

Evidence so far: literature, standards and industry reports only; no human participants have been involved yet.

Appendix G: Full functional and non-functional requirements

ID	Priority	Requirement	Verification	Trace
FR01	Must	Record sample ID and metadata	Data audit	O2
FR02	Must	Acquire spectra with white and dark references, three scans	File inspection	O2
FR03	Must	Triplicate oven moisture with timing	Mass log	O2, O6
FR04	Should	Retain sealed subsample	Sample log	O8
FR05	Must	Configurable preprocessing inside CV	Unit and leakage tests	O3
FR06	Must	PCA, T2 and Q outlier detection	QC report	O2
FR07	Must	Lock and hash test set	Hash check	O6
FR08	Must	CARS and comparator selection	Run log	O4
FR09	Must	PLSR, SVR, RF on identical splits	Run log	O5
FR10	Could	Neural model if about 150+ samples	Run log	O5
FR11	Must	Report RMSEC, RMSECV, RMSEP, R2, RPD, bias	Report	O6
FR12	Must	Model registry per parameter	Code review	O8
FR13	Must	Predict with interval and range flag	Out-of-range test	O7
FR14	Should	Log time and predictions	Log review	O6
FR15	Could	Train additional parameter models when reference data arrive	Run log	O8
FR16	Could	Streamlit interface	Demo	O7
NFR01	Must	R2p at least 0.90, RPD at least 2.5	Locked test	H1
NFR02	Must	Under 60 s per sample	Timing	H4
NFR03	Should	Repeat-scan SD at most 0.2%	Repeat scans	O7
NFR04	Must	Reproducible from code, seeds, data version	Clean re-run	Integrity
NFR05	Must	Runs on laptop CPU	Field use	O7
NFR06	Should	SUS at least 68	Questionnaire	O7

ID	Priority	Requirement	Verification	Trace
NFR07	Must	No personal data; coded suppliers; two backups	Audit	Ch. 4
NFR08	Must	Screening label on outputs	UI review	Ch. 4

Appendix H: Use cases, use-case descriptions and activity/sequence diagrams

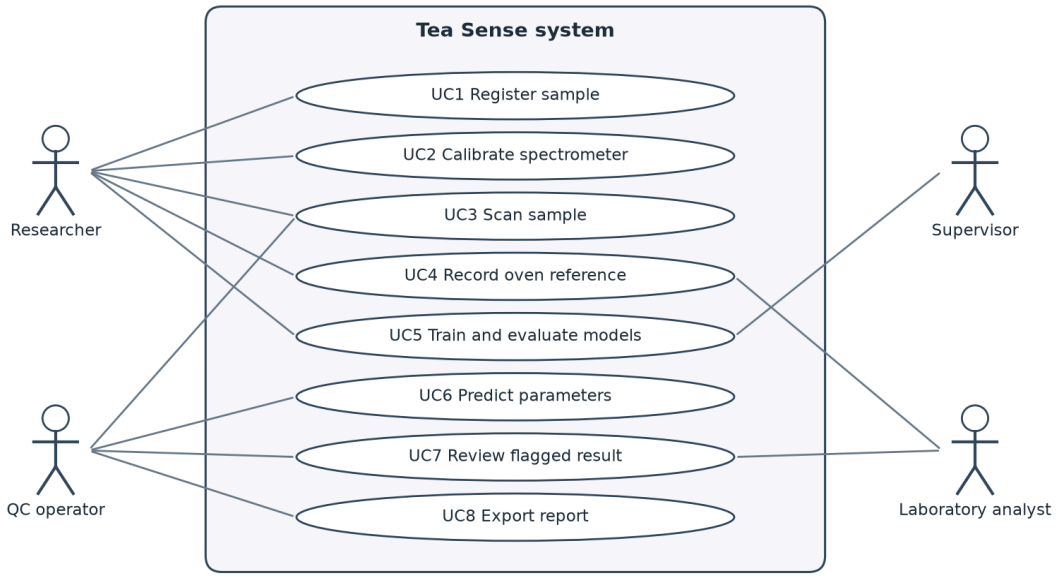


Figure 6: Use-case diagram

Use case	Actor	Main flow	Alternative flow
UC3 Scan sample	Researcher, QC operator	Load sample in holder; scan three times with repacking; average; save	Reference expired: re-run white and dark references
UC4 Record oven reference	Researcher, lab analyst	Weigh 3 to 5 g in triplicate; dry at 103 C; cool; weigh to constant mass; log time	Replicate SD above 0.2%: repeat
UC6 Predict parameters	QC operator	Load spectrum; apply registry; show value, interval, flag, label	Out of range: show "send to laboratory"
UC7 Review flagged result	QC operator, lab analyst	Send flagged sample for laboratory test; record result	Repeated flags: report to researcher for recalibration

Activity (predict): start, calibrate, scan x3, average, preprocess, out-of-range check, then either predict and display or flag for laboratory, end.

Appendix I: Detailed architecture and data-flow diagrams

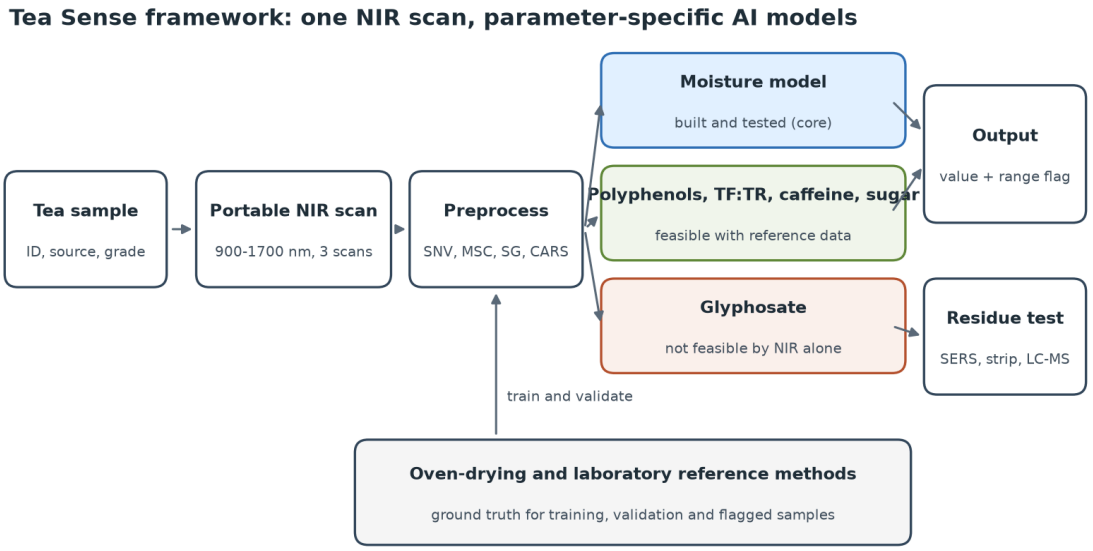


Figure 7: Conceptual architecture (repeat of Figure 2 for reference)

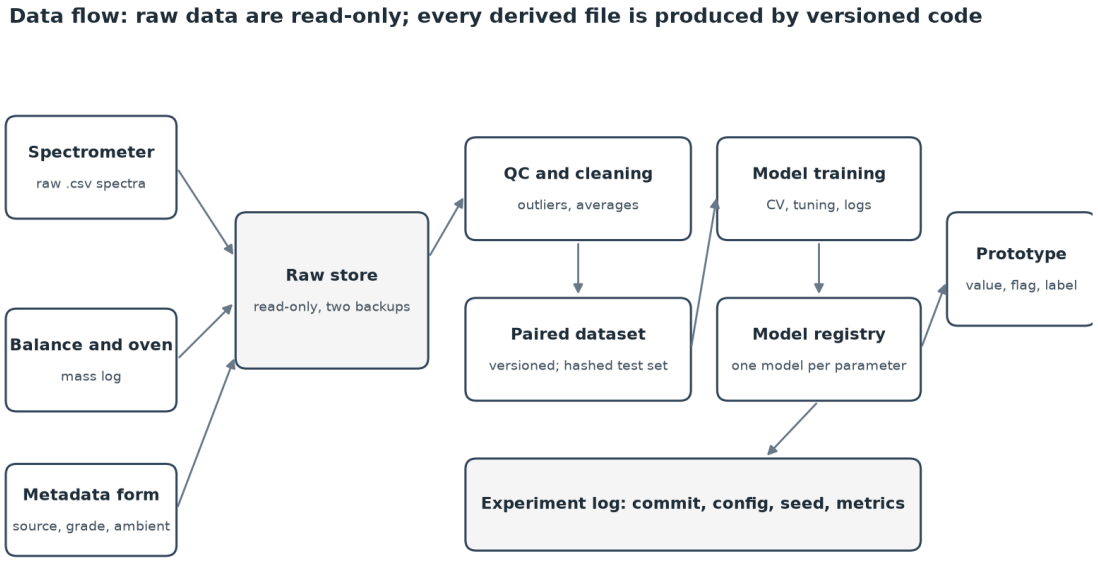


Figure 8: Data-flow diagram

Appendix J: Detailed Gantt chart, backlog and milestone plan

Phase	Tasks	Start	End	Milestone
P1 Research and proposal	Review; matrix; proposal	Sep 2026	Oct 2026	Proposal submitted
P2 Ethics, protocol, pilot	Ethics form; SOPs; pilot	Nov 2026	Nov 2026	Approval
P3 Data collection	Source; scan; oven-dry; retain	Dec 2026	Feb 2027	Dataset v1
P4 Cleaning and baseline	QC; split; PLSR baseline	Jan 2027	Feb 2027	Test set locked
P5 Multi-parameter synthesis	Evidence grading; registry design; partner outreach	Jan 2027	Mar 2027	Feasibility matrix
P6 Preprocessing and selection	Grid; CARS; SPA/iPLS	Feb 2027	Mar 2027	RQ2 tables
P7 Models	SVR; RF; optional NN; tuning	Mar 2027	Mar 2027	Tuned models
P8 Evaluation and prototype	Locked test; LOSO; timing; prototype	Mar 2027	Apr 2027	RQ1, RQ3
P9 Field feasibility	Field trial; oven subset; SUS	Apr 2027	Apr 2027	RQ4
P10 Write-up and viva	Chapters; proofreading; viva practice	Apr 2027	May 2027	Final thesis

Backlog priority rule: Must items first; Could items (neural model, Streamlit, partner-lab pilot) only after the moisture evaluation is complete.

Appendix K: Full risk register and issue log

Risk	L	I	Mitigation	Contingency
Spectrometer unavailable	M	H	Confirm early; second source	Benchtop at partner lab
Too few samples or narrow range	M	H	Several suppliers; conditioning	Accept 80; drop NN
Inconsistent oven values	M	H	SOP; triplicates; pilot	Re-run; exclude
Moisture change before drying	M	M	Sealed bags; short window	Record time gap
Weak model performance	M	M	Systematic search; QC	Report as valid finding
Data leakage	M	H	Locked test; pipelines	Re-split; disclose
No partner lab	H	M	Early outreach	Evidence-only (planned)
Scope creep	H	H	MoSCoW	Future work
Field conditions	H	M	Log conditions	Report as finding
Instrument drift	M	M	References; control sample	Drift correction
Timeline slip	M	M	Buffer; weekly review	Drop Could items
Data loss	L	H	Git; two backups	Restore
Limited chemometrics skills	M	M	Learning plan; supervisor	Simplify to PLSR focus
Ethics delay	L	M	Submit early	Start with retail samples

Issue ID	Date raised	Description	Action	Status
I-01				Open
I-02				

Appendix L: Generative AI use declaration and AI-use register

The declaration is in the front matter and Section 4.6.6.

Tool	Date	Purpose	Input category	Verification	Outcome
ChatGPT / other (author to complete)	Aug 2026	Language refinement of Chapter 1 draft	Author's own draft text	Author review	Retained after editing
Claude (Anthropic)	19 Oct 2026	Structuring proposal to template; drafting assistance for Chapters 1 to 4 and appendices	Author's draft, survey sheet, public papers	Author to review and rewrite every section	To be rewritten or retained by author
Claude (Anthropic)	19 Oct 2026	Literature-search support for caffeine, TF:TR, polyphenols, sugar, glyphosate	Public search results	DOIs checked via Crossref; author to read papers	Retained; marked NEW
Claude (Anthropic)	19 Oct 2026	Preliminary proof-of-concept code on public Tecator data	Public dataset	Code run and outputs inspected; author to re-run	Retained as preliminary

Appendix M: Preliminary proof-of-concept screenshots, test logs and sample outputs

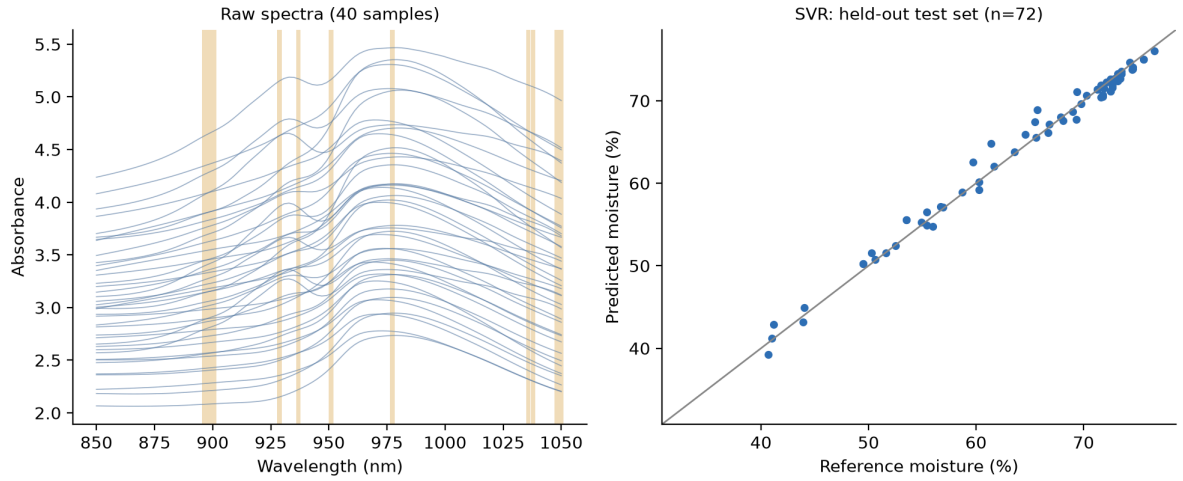


Figure 9: Proof of concept on surrogate Tecator data: raw spectra with CARS-selected wavelengths shaded (left) and SVR predictions on the locked test set (right)

Preprocessing (PLSR)	Latent variables	RMSECV (%)	RMSEP (%)
Raw	11	2.761	2.276
SNV	10	1.955	1.921
MSC	11	2.283	2.307
SG smoothing	11	2.817	2.488
SG 1st derivative	11	2.781	2.542
SNV + SG 1st derivative	10	1.949	1.921
SG 2nd derivative	5	2.66	2.769

Log summary. Data: OpenML dataset 505 (Tecator), 240 samples, 100 absorbances 850 to 1050 nm, moisture 32.8 to 76.6%. Split: Kennard-Stone 168/72. CV: 10-fold, seed 42. Best preprocessing by RMSECV: SNV + SG 1st derivative. CARS: 10 repeats of 50 iterations; best subset 11 wavelengths, 10 latent variables. SVR best parameters: C 1000, gamma 0.001, epsilon 0.2. Paired Wilcoxon SVR vs PLSR absolute errors: p = 1.2e-5. Prediction time 1.46 ms per sample (computation only). Script: poc.py (Python 3, scikit-learn 1.9).

Appendix N: Ethics clearance material and data-management approvals

Status: to be attached after approval. Planned contents: ethics application form; participant information sheet and consent form for field-test users; supplier permission letters; laboratory risk assessment for oven, glassware and desiccant use; data-management plan approval.

Appendix C (continued): Literature survey matrix

Citation	Problem	Data	Method	Evaluation	Findings	Limitations	Relevance
Hazarika et al. (2017), Two and a Bud	Need for a rapid, reagent-free alternative to chemical and sensory tea-quality testing	Initial experiments at Tocklai TRI and Jadavpur University; no named dataset	Review of NIR (780 to 2500 nm) diffuse reflectance with chemometric calibration	Narrative review	NIR is a green, fast, non-destructive tool for tea quality and process control	Review; NIR's own technical limits not detailed	High: foundational justification for NIR in tea
Das et al. (2020), ANS2020	Discriminating green teas by gallic acid content	Five green-tea variants; HPLC reference	NIR reflectance 900 to 1700 nm; preprocessing; PCA; PLSR	Class Separability Index	Preprocessing raised separability from 53.63 to 93.20	Abstract-level detail; gallic acid not moisture	High: shows preprocessing effect (RQ2)
Revilla et al. (2024), Foods 13(3) 450	Predicting heavy metals and blend percentages in tea	322 samples: 16 pure black, green and red teas plus lab blends; ICP-MS reference	Foss NIRSystem 5000 (1000 to 2498 nm); MPLS, PCA, SIMCA, OPLS-DA	Calibration vs validation accuracy	98.3% calibration but 92.3% validation accuracy on some blends	Metals have no direct NIR peak; small sub-categories	High: calibration overstates accuracy (Gap 3)
Hazarika et al. (2024)	Rapid valuation of CTC black tea (liquor scores and theaflavin)	100 commercial CTC samples from Kolkata and Upper Assam	Portable InGaAs NIR 900 to 1700 nm; SNV, MSC, 1D, 2D; PLSR with CARS and Random Frog	Prediction statistics vs spectrophoto meter and sensory panel	Few selected wavelengths gave strong models	Sensory reference is subjective; moisture not targeted	High: black tea, portable NIR, CARS and PLSR template

Citation	Problem	Data	Method	Evaluation	Findings	Limitations	Relevance
Huang et al. (2024), Foods 13(3) 389	Identifying coffee roast degree and variety	Three varieties, four roast levels; 125 tablets per sample	THz-TDS 0.06 to 4.5 THz; FFT; PCA; LD, SVM, NB, KNN	Classification accuracy	Linear discriminant classified best	THz blocked by water; samples dried below 5% moisture	Medium: shows why THz is unsuitable for moisture
Qiu et al. (2026), Agric. Products Processing and Storage 2:22	State of spectroscopic quality detection across agricultural products	Review of many studies (fruit, meat, aquatic products)	Surveys HSI, Vis-NIR, NIRS, MIRS, Raman, THz and others; SG, SNV, MSC; SPA, CARS, GA; PLSR, SVM, ANN, CNN	Narrative synthesis	Identifies poor universality, environmental interference, cost and data volume as bottlenecks	Cross-product review	High: frames validation and field challenges (Gap 3, Gap 4)
Jin et al. (2024), Gems and Gemology 60(4)	UV-Vis-NIR spectroscopy for gem identification	Illustrative gem spectra	Absorption and reflectance spectroscopy; Beer-Lambert law	Qualitative interpretation	Explains chromophores and instrument set-ups	Not food or tea	Excluded: off-domain
Scheibelhofer et al. (2016), manuscript	Effect of particle size and density on NIR reflectance of powders	Seven lactose grades	Spatially resolved NIR 900 to 1700 nm; Monte Carlo photon simulation	Simulated vs measured absorbance	Particle size and packing change the sampled volume and spectra	Simplified particle shapes; DOI not final	Medium: supports repacking and scatter correction for tea particles

Citation	Problem	Data	Method	Evaluation	Findings	Limitations	Relevance
Martini and Arias (2021), J. Integrative Neuroscience	NIR spectroscopy and stimulation in cognitive neuroscience	No original data	Commentary	None	Discusses fNIRS and photobiomodulation	Medical, not food	Excluded: off-domain
Picton et al. / Germon et al., BJA correspondence	Reliability of NIR cerebral oximetry	Prior clinical studies	Correspondence	None	Debate on extracranial contamination	Medical, not food	Excluded: off-domain
Beese et al. (1998), Stroke 29(10)	NIRS vs evoked potentials for cerebral ischaemia	317 surgical patients	INVOS 3100A oximeter; statistical tests	Clinical comparison	Clinical NIRS detection limits	Medical, not food	Excluded: off-domain
Zhou et al. (2026), Beverages 12(3) 32	Tea variety traceability	84 spectra: four varieties, seven samples each, three scans	Bruker FT-NIR 830 to 2500 nm; SG, MSC, SNV, 1D; PCA, LDA, SPA, SVM	Classification accuracy	NIR with preprocessing separates varieties	Four varieties; small set; limited external validation	Medium: preprocessing choices; small-sample caution
Damayanthi et al. (2026), ISAE 2026	Detecting sugar adulteration in Sri Lankan black tea	250 samples from seven Sri Lankan regions; 100 analysed	UV-Vis 190 to 1100 nm; SG, PCA, PLS-DA, RBF-SVM; 70:30 split	Classification on test split	Chemometrics and ML detect added sugar	Abstract only; limited validation detail	Medium: local precedent for chemometrics on Ceylon tea

Citation	Problem	Data	Method	Evaluation	Findings	Limitations	Relevance
Pan et al. (2026), BIO Web of Conferences 230	Estimating tea plant nitrogen status	Tea garden canopy spectra with lab nitrogen	Derivative spectroscopy; spectral indices; PCA; linear regression	Regression fit	Derivatives help extract nitrogen signal	Field canopy effects; limited generalisation	Low: derivative preprocessing evidence only
Zhao et al. (n.d.), preprint	Classifying teas and modelling fermentation	38 green and oolong specimens plus Pu-erh samples	LIBS; SG; VIP; Random Forest	Classification accuracy	LIBS fingerprints separate tea types	Destructive; small set	Low: LIBS rejected for moisture
Li et al. (2023), Food Chemistry: X 20	Grading Keemun black tea	Seven grades, 30 samples per grade; nine-person sensory panel	Smartphone micro-NIR 900 to 1700 nm, vision, colorimetric array; SVM, LS-SVM, ELM; data fusion	Grade classification and regression	Fusion improved grading	One supplier; grinding needed	Medium: phone-linked NIR is practical for black tea
Damayanthi et al. (2025), RESCON 2025	Sugar adulteration in Sri Lankan black tea by FTIR	Orthodox black tea from seven regions; 1 to 40% sugar	FTIR-ATR; SNV, MSC, derivative; PCA, OPLS-DA	Classification	FTIR with chemometrics detects added sugar	Abstract only; sample counts not given	Medium: local Sri Lankan evidence
Liu et al. (2020), J. Algorithms and Computational Technology	Reverse design of RFID tag antennas	Simulated antenna images	Image analysis; HFSS simulation	Simulation	Improves antenna matching	Not tea or spectroscopy	Excluded: off-domain

Citation	Problem	Data	Method	Evaluation	Findings	Limitations	Relevance
An et al. (2025), Foods 14(9) 1442	Predicting moisture during black-tea withering	Jiukeng leaves, 12-hour controlled withering, hourly samples; moisture analyser reference	IAS-3100 NIR 900 to 1700 nm plus machine vision; VISSA; SVR; low, mid, high-level fusion	Regression metrics on validation	Fusion and VISSA improved moisture prediction	One cultivar, one harvest, controlled withering	High: black-tea moisture, wavelength selection
Zou et al. (2025), Foods 14(16) 2829	Predicting tea pigments during rolling	150 rolling samples of Fuding Dabai	Portable NIR plus vision; PLSR, SVR, LS-SVR; wavelength selection; fusion	Regression metrics	Fusion predicted theaflavins and thearubigins	One cultivar, controlled processing	Medium: future extension parameter
Xu et al. (2023), Applied Sciences 13 1739	Polyphenols in fresh tea leaves	800 leaves of two cultivars, one garden, one date	Dual-band Vis/NIR device; SG; Random Frog; LS-SVR; spectral fusion	Regression metrics	Fused bands improved polyphenol prediction; portable device built	Single site and date	Medium: device design and future polyphenol work
Zou, Li and Han (2024), Agronomy 14 1582	Quality of Sichuan dark tea	56 samples, three scans each	Thermo Antaris II FT-NIR; SNV, MSC, derivatives; siPLS; fluorescence EEM; XGBoost	Regression and classification	NIR predicted constituents of dark tea	Two tea types, local markets	Low to medium: preprocessing and siPLS evidence

Citation	Problem	Data	Method	Evaluation	Findings	Limitations	Relevance
Piyasena et al. (2026), Bull. National Research Centre 50	Seasonal variation in polyphenols of Sri Lankan cultivars	TRI 2025 and DT-1 over 22 months at St. Coombs Estate	Folin-Ciocalteu assay; oven, freeze and fluid-bed drying	Statistical comparison	Polyphenols vary by month, climate and drying	Two cultivars, one estate; chemical not NIR	Medium: local reference chemistry; seasonal variation risk
Piyasena et al. (2022), Applied Food Research 2	Natural sugars in tea leaves and black tea	Sri Lankan samples incl. 90 low-grown black teas	HPLC with RI detector	Statistical comparison	Sugars vary by maturity, cultivar and process	Needs extraction; not spectroscopy	Low: background on local reference methods
Sarma and Dashora (2026), Sustainable Food Technology 4	Predicting sensory quality of tea blends	30 blend runs of four Assam teas; semi-trained panel	Biochemical assays; ML classifiers; five-fold CV; SHAP	Cross-validated accuracy	Interpretable ML links chemistry to sensory scores	Assam only; subjective labels; no NIR	Low: ML validation practice
Zhang et al. (2023), Sensors 23(24) 9684	Organic matter in tea-garden soil	150 soil samples from three gardens	Hyperspectral imaging 300 to 1000 nm; SNV, MSC; Random Frog, VCPA-IRIV; SVR vs PLSR; fusion	Regression metrics	Fusion and selection improved prediction	One province; lab conditions	Low: selection and SVR vs PLSR evidence

Citation	Problem	Data	Method	Evaluation	Findings	Limitations	Relevance
Huang et al. (2022), Frontiers in Physics 10	Moisture in cattle feed	108 samples, 29.46 to 49.46% moisture; 2:1 split	THz-TDS plus handheld NIR 900 to 1700 nm; derivative, SG; CARS feature fusion; PLSR	Calibration and validation metrics	CARS-based fusion improved moisture prediction	Not tea; THz needs pellets and dry air	High: CARS and PLSR for moisture
Lan et al. (2022), Foods 11(18) 2928	Moisture during green-tea fixation	Zhongcha 108; one site; samples every 25 s	NIR-S-R2 micro-NIR plus vision; SNV; PLSR, SVR, Elman, WOA-Elman; fusion	Regression metrics	Optimised Elman with fusion beat PLSR and SVR	One cultivar, site and run	High: micro-NIR tracks tea moisture; PLSR vs non-linear
Guo et al. (2025), Food Chemistry: X 27	Origin traceability of Rizhao green tea	105 samples from six regions; three tasters	IAS3100 NIR plus hyperspectral; MSC, z-score; PCA, SVM, RF, CNN; PLSR, SVR	Classification and regression	Fused features gave high origin accuracy	Modest dataset; overoptimism risk	Medium: multi-region validation lessons
Andasuryani et al. (2024), IOP Conf. Series EES 1426	Moisture in Gambir herbal tea powder	60 samples; 40 calibration, 20 validation; oven reference	Buchi NIRFlex N-500 1000 to 2500 nm; PLS	Calibration and external validation	PLS predicted moisture with oven-drying reference	One variety; small set; not Camellia sinensis	High: simple PLS plus oven template

Citation	Problem	Data	Method	Evaluation	Findings	Limitations	Relevance
Tao et al. (2026), Scientific Reports 16	Real-time moisture in white-tea withering	17,856 line observations; chronological split	MS-580 NIR analyser above conveyor; MSC; mRMR; attention BiGRU; XGBoost	Chronological train, validation, two test sets	Deep model predicted moisture online	One region and season; white tea	High: in-line feasibility and leakage-safe splits
Kang et al. (2026), J. Food Measurement and Characterization 20	Tea moisture monitoring during withering	215 batches, 1,505 spectra, 0 to 80% moisture	ATP8600 NIR; adaptive spectral sampling; TSNet (GRU, attention, CNN)	Lab and online validation	Deep model with adaptive sampling monitored moisture online	Controlled lab calibration; transfer unverified	High: deep models need large, wide-range data
Zou et al. (2022), Horticulturae 8(12) 1170	Moisture of black tea across processing	305 spectra from Jiukeng across fresh leaf, withering, rolling, fermentation, drying; oven reference	Micro-NIR 900 to 1700 nm; SNV, MSC, SG, z-score; PCA; PLS vs GWO-SVR	Calibration and prediction metrics	GWO-tuned SVR beat PLS	Single cultivar; controlled conditions	Very high: closest design to Tea Sense
Zhang, Gao and Guo (2026), Applied Sciences 16(3) 1614	Year-to-year quality variation of Shiqian Taicha	116 commercial batches 2021 to 2023	Antaris II FT-NIR calibrated predictions; fuzzy membership scoring	Comparison by year and tea type	NIR moisture and constituents used in routine monitoring	Commercial data; climate confounding	Medium: NIR moisture in routine practice

Citation	Problem	Data	Method	Evaluation	Findings	Limitations	Relevance
NEW: Rehman et al. (2020), Foods 9(6) 827	Rapid caffeine measurement in tea as a complement to HPLC	25 commercial teas (Oman); 45 caffeine standards spiked into tea solution	FT-NIR 4000 to 10000 cm-1; unit vector normalisation; PLSR (3 factors); 75:25 split	RMSECV, external RMSEP, LOQ	Test RMSEP 0.08 ppm, r2 0.97, LOQ 0.3 ppm; agreed with HPLC (15 min run per sample)	Liquid tea solutions, not leaf; inconsistent figures between text and table	High: caffeine evidence (RQ4)
NEW: Zareef et al. (2018), Analytical Methods 10(25)	Predicting amino acids, caffeine, theaflavins and water extract in black tea	Black tea samples (count to verify)	FT-NIR; PLS, Si-PLS, GA-PLS, Bi-PLS	Calibration and prediction metrics	Bi-PLS gave best caffeine and theaflavin models with fewer variables	Full text not accessed; sample count unverified	High: black-tea caffeine and TF in one spectrum
NEW: Dong et al. (2017), Food Sci. Biotechnol. 26(4)	Fermentation quality indices of black tea	Black tea fermentation samples	NIR; Si-PLS interval selection; ELM boosted by AdaBoost	RMSEP, RPD, R per index	Predicted catechins, TP/FAA and TF/TR with RPD above 2.5 and R above 0.9	One processing setting	Very high: direct TF:TR ratio evidence
NEW: Chen et al. (2018), J. Food Sci. Technol. 55(10)	Taste quality and taste compounds of black tea	Black tea samples	FT-NIR; Si-PLS; BP-AdaBoost	Prediction metrics	Simultaneous estimation of taste quality and related compounds	Full text not accessed	Medium: multi-parame ter from one spectrum

Citation	Problem	Data	Method	Evaluation	Findings	Limitations	Relevance
NEW: Bai et al. (2022), Scientific Reports 12, 3833	Quality parameters of instant tea	118 instant tea samples	NIR; SVR, BPSO-SVR, PLS, BPSO-PLS	Validation Rp and RMSEP	Moisture Rp 0.971, caffeine 0.976, polyphenols 0.795, polysaccharides 0.819	Instant tea, not leaf; polyphenol model weaker	High: one spectrum, four parameters; shows uneven feasibility
NEW: Liu et al. (2024), Sensors 24(11) 3362	Four catechins in black tea	Black tea samples (count to verify)	NIR; FIC-SS wavelength selection; ELM vs PLS	Prediction metrics	ELM beat PLS for EC, ECG, EGC and EGCG	Overlapping catechin bands; small-sample problem	High: polyphenol sub-components and non-linear models
NEW: Xu et al. (2026), RSC Advances 16(3)	Polyphenols-to-amino-acids ratio in Wuyi Rock Tea	Wuyi Rock Tea samples	NIR; SNV plus SG derivative; CARS-PLSR	R2c, R2p, RPD	R2p 0.98 and RPD 3.21 for the TP/FAA ratio with CARS-PLSR	Oolong-type tea; one region	High: ratio targets and CARS
NEW: Luo et al. (2023), Spectrosc. Spectral Anal. 43(8) 2649	Added sucrose in finished black tea	Fengqing black tea with 0, 250, 500, 750 g added sucrose	NIR; SNV; CARS, SFLA, VCPA-IRIV, VISSA; PLSR, SVR, RF	Rc, Rp, RPD	VCPA-IRIV-SVR best: Rp 0.924, RPD 2.51; non-linear beat PLSR	Abstract-level detail; one tea type	Very high: direct NIR sugar-in-black-tea evidence

Citation	Problem	Data	Method	Evaluation	Findings	Limitations	Relevance
NEW: Tang et al. (2025), Foods 14(24) 4261	Trace colourant adulteration in black tea	345 spectra; colourants at 0.1 to 0.5 g/kg	Handheld NIR-R210 900 to 1700 nm; SMAFNet deep network vs PLS-DA, SVM, RF, CNN	Validation accuracy	97.2 to 100% accuracy	Lab-spiked powder; one tea	High: handheld NIR plus deep learning detects g/kg adulterants
NEW: Roberts and Smith (1961), The Analyst 86, 94	Measuring theaflavins and thearubigins in black tea liquor	Black tea liquors	Solvent extraction and spectrophotometry	Comparison with tasters	Established the TF and TR reference method	Needs solvents and laboratory work	High: reference method for TF:TR
NEW: Samanthy et al. (2021), iPURSE 2021, p.158	Glyphosate residues in tea	Fortified tea samples	Water/chloroform extraction; LC-MS/MS with MRM	Recovery, precision, LOD/LOQ	LOD 1.1 and LOQ 1.8 mg/kg; recoveries 92 to 100%	Specialist LC-MS/MS laboratory	High: shows glyphosate is a trace (mg/kg) target